

Sensitivity Analysis for Nonlinear Models with a Synthetic Confounder.

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Abstract

This paper propose a sensitivity analysis approach for general likelihood-based models. A synthetic confounder is constructed via simulation which has pre-specified partial correlations with the residualised treatment and outcome variables, thus capturing researchers' beliefs about the strength of the confounding. By including the synthetic confounder in the short (misspecified) model, the framework yields exact adjusted average marginal effects. Further, it is possible to calculate fragility thresholds via a bisection algorithm and includes a benchmarking procedure to calibrate the sensitivity parameters against observed covariates. A Monte Carlo simulation shows that the new framework outperforms direct application of existing sensitivity approaches for linear models under moderate to strong confounding. The new framework is applied the 1970 British Cohort Study to evaluate the sensitivity of the effect of early life cognitive skills on labour market participation and job quality during adulthood.

Key Words: *Sensitivity analysis, omitted variable bias, nonlinear models, average marginal effects, synthetic confounder, fragility thresholds*

JEL Classification: *C25, C35, C52, C18.*

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1 Introduction

The omission of relevant variables is one of the main challenges in economics and statistics. Excluding a variable that is correlated with both the outcome and a predictor of interest (e.g. a treatment indicator) causes the estimated coefficient of the predictor to absorb part of the omitted variable’s influence, producing a bias. In particular, the resulting estimates cannot be given a causal interpretation because they confound the effect of the predictor with the effect of the omitted variable or confounder, and policy conclusions drawn from them may be misleading.

Starting with the seminal contributions of Cornfield et al. (1959); Bross (1966, 1967) and Rosenbaum and Rubin (1983), numerous methods have been developed to assess how sensitive the conclusions of observational studies are to the possible presence of such confounders (Altonji et al., 2005; Oster, 2019; VanderWeele and Arah, 2011; Ding and VanderWeele, 2016; Carnegie et al., 2016; Blackwell, 2013; Frank, 2000; Cinelli and Hazlett, 2020; Masten and Poirier, 2022; Sjölander et al., 2026; Brumback et al., 2004, among others). These methods explore how strong would an omitted confounder need to be to overturn the result of a study. If moderate confounding is sufficient to overturn the findings, then the conclusions of the study are fragile and should be treated with caution regardless of their statistical significance.

A large proportion of the results in the sensitivity literature have focused on linear models. Not only do they constitute the workhorse of modern empirical practice, but they also often allow for closed form sensitivity formulas with known properties (e.g. Oster, 2019; Cinelli and Hazlett, 2020). However, researchers often need to deal with discrete outcomes or outcomes whose analysis benefit from the use of nonlinear models for precision and efficiency reasons. The question is whether existing linear methods can be applied to these nonlinear models (effectively using a linear probability approximation) and, if not, which alternative approaches can be considered¹.

This paper explores both questions by studying how much Average Marginal Effects

¹This is a general question which does not depend on whether the researcher is willing to maintain a selection-on-observables assumption (with the intention of establishing a causal claim).

(AME) from a likelihood-based nonlinear model would change if there were an omitted variable with given partial correlations with the treatment and outcome residuals. The paper makes three principal contributions. First, it documents the omitted variable bias of general likelihood-based models. It is shown that the expression of the omitted variable bias for general likelihood-based methods admits (to first order) the interpretation of the slope coefficient of a linear regression. This suggests that the framework in Cinelli and Hazlett (2020) could be directly applied to nonlinear models replacing OLS residuals with, for example, standardised residuals. However, the bias includes a remainder term that reflects the curvature of the log-likelihood and, more importantly, grows quadratically with the strength of the confounder. This second order remainder term implies that direct application of linear methods in nonlinear settings is valid as first-order approximations but only when confounding is weak.

The above discussion implies that there is scope for the development of bespoke methods for nonlinear models. Therefore the paper proposes a new simulation based approach to sensitivity analysis for omitted variable bias in nonlinear models. The method is directly inspired by Cinelli and Hazlett (2020), who derive exact expressions for the omitted variable bias in linear models as a function of the partial correlations of the omitted variable with the residualised treatment and outcome. The paper exploits the linear index structure common to likelihood-based models to construct a synthetic confounder with specified partial correlations with the treatment residual and the outcome residual. By augmenting the short (misspecified) model with this synthetic confounder and re-estimating by maximum likelihood, the method delivers the adjusted average marginal effect at each point in the sensitivity parameter space, without requiring the confounding to be weak. The framework applies to likelihood-based models, including logit, probit, complementary log-log, Poisson, negative binomial, ordered logit, and multinomial logit.

The sensitivity parameters in the synthetic confounder approach are partial correlations between the synthetic confounder and the residualised treatment and outcome respectively. Therefore, they are bounded in $[-1, 1]$, simple to interpret, and directly comparable across models and datasets. Following Cinelli and Hazlett (2020) and Masten

and Poirier (2022), the framework enables the calculation of explain-away and sign-change thresholds for every value of ρ_Y in the grid. This is done through a grid-search plus bisection algorithm. As in Cinelli and Hazlett (2020), the paper develops a benchmarking procedure that calibrates the sensitivity parameters against observed covariates, grounding the analysis in the specific empirical setting.

The new method addresses the three main simplifying assumptions identified by Ding and VanderWeele (2016). The synthetic confounder Z is constructed as a continuous variable whose distribution is determined by the data through the residualised treatment and outcome, with an independent noise component absorbing the remaining variation needed for unit variance. No distributional restriction is imposed on Z beyond this. In particular, Z need not be binary. Second, while the discussion assumes homogeneity² for expositional convenience, this assumption is not required. Third, the method performs sensitivity analysis under the full range of alternative hypotheses rather than only under the null of no effect by delivering an exact adjusted average marginal effect at every feasible (ρ_D, ρ_Y) point in the sensitivity parameter space. Furthermore, the asymptote problem identified by Masten and Poirier (2022) for the δ parameter in Oster (2019) does not arise in our framework because (ρ_D, ρ_Y) are bounded partial correlations lying in a compact set and therefore large sign-change thresholds are possible without the denominator collapse that causes Oster’s δ to diverge.

Having developed the synthetic confounder approach, the paper then applies the method to evaluate the sensitivity of cognitive skills as a predictor of labour market participation and job quality. More precisely, using data from the 1970 British Cohort Study (BCS70), we estimate short models of full-time employment and managerial or supervisory status at age 46 as a function of a composite measure of cognitive ability assessed at age 10, controlling for a rich set of early-life covariates including perinatal characteristics, parental background, and teacher-reported measures of developmental behaviour. Because cognitive skills are measured well before the adult outcomes of interest are realised, the design reduces concerns about reverse causality. However, unobserved variables that

²No interaction between the confounder and the treatment in the linear index

are positively associated with both childhood cognitive skill and adult labour-market success (such as family socioeconomic advantage or parental ability) would cause the short model to overstate the causal effect of cognition. In the absence of a sensitivity analysis, the researcher must implicitly invoke a conditional independence assumption, maintaining that the observed controls are sufficient to absorb all such confounding. The synthetic confounder method allows us to explore how strongly correlated with both cognitive skill and labour-market outcomes would an omitted variable need to be before the conclusions change.

This is not the first paper to propose sensitivity methods for nonlinear models. Indeed, by working within a potential outcomes framework, the literature has left distributional assumptions implicit. For example, Imbens (2003); Brumback et al. (2004); Blackwell (2013) work within a potential outcomes framework that imposes no distributional assumption on the outcome at the identification level. However, Imbens requires a normal linear structure to derive closed-form sensitivity contours after integrating out the unobserved confounder, while the outcome adjustment in Blackwell is additive and its R^2 reparameterisation relies on a linear variance decomposition, neither of which extends naturally to likelihood-based nonlinear models. Brumback et al. (2004) develop a sensitivity analysis framework designed for time-varying treatments estimated by inverse probability weighting and does not extend naturally to the likelihood-based cross-sectional models. In particular, their framework does not deliver adjusted average marginal effects at each sensitivity parameter value.

VanderWeele and Arah (2011) derive bounds for the bias due to unmeasured confounding for general outcomes within a potential outcomes framework, under a homogeneity assumption that rules out interactions between the confounder and treatment. Ding and VanderWeele (2016) introduce the E-value, defined as the minimum strength of association on the risk ratio scale that an unmeasured confounder would need to have with both treatment and outcome to fully explain away an observed association; its applicability is primarily to binary outcomes and exposures. Recently, Sjölander et al. (2026) extend the E-value to marginal causal effects, however their framework retains binary outcomes

and exposures and it produces bounds on the causal effect rather than adjusted point estimates.

The paper closest to ours is Carnegie et al. (2016), who introduce a simulated potential confounder approach in which a synthetic confounder is drawn from the conditional distribution implied by a specified complete-data likelihood, for pre-specified values of the confounder’s coefficient in the treatment and outcome models. When both the outcome and treatment are continuous and normally distributed, this conditional distribution is available in closed form. For binary treatment they employ a stochastic EM algorithm. Park et al. (2026) extend the simulation-based sensitivity analysis of Carnegie et al. (2016) to causal decomposition analysis with optimal treatment regimes, and develop a formal benchmarking strategy for binary treatments using original data scales. The method presented here differs from theirs in four respects. First, the method is parameterised in terms of partial correlations rather than coefficients, making the sensitivity parameters scale-free and directly comparable across models. Second, the approach defines a closed-form feasibility constraint on achievable parameter pairs. The method supports the full range of GLMs used in applied economics, including Poisson, negative binomial, ordered logit, and multinomial logit, which Carnegie et al. do not handle. Finally, a bisection algorithm is implemented to compute fragility thresholds analytically rather than reading them from a contour plot.

The remainder of the paper proceeds as follows. Section 2 derives the approximate bias formula for likelihood-based models, revealing how second-order curvature terms invalidate linear approximations under strong confounding. Section 3 develops the simulation-based sensitivity analysis framework. Section 4 introduces fragility thresholds and benchmarking procedures that allow researchers to calibrate sensitivity parameters against observed covariates. Section 5 presents a Monte Carlo simulation to evaluate the method. Section 6 contains the empirical application, while section 7 concludes.

2 Omitted Variable Bias.

Assume the existence of a sample of $i = 1, \dots, n$ observations. Let Y_i denote an outcome variable with range depending on the model, (e.g. $Y_i \in \{0, 1\}$ for binary models, $Y_i \in [0, 1]$ for fractional models, $Y_i \in \{1, \dots, J\}$ for multinomial models, etc). Let $D_i \in \mathbb{R}$ be the variable of interest (e.g. a treatment indicator or a continuous measure of income or cognitive skills). In what follows, to fix ideas, we give D the interpretation of a treatment variable in the causal inference sense. $\mathbf{X}_i \in \mathbb{R}^p$ is a vector of observed covariates. The variable $Z_i \in \mathbb{R}$ is an unobserved confounder. Define the observed regressor vector $W_i^* = (A_i', D_i)' = (1, \mathbf{X}_i', D_i)'$ and the full regressor vector $W_i = (A_i', D_i, Z_i)' = (1, \mathbf{X}_i', D_i, Z_i)'$. Here $A_i = (1, \mathbf{X}_i)'$ constitute a set of nuisance variables which are not of primary target of the analysis. Let $f(y | w; \boldsymbol{\theta})$ denote the model density or mass function for outcome Y_i given regressors W_i under parameter $\boldsymbol{\theta} = (\boldsymbol{\theta}_A', \beta, \delta)'$, where β is the parameter associated with the variable of interest (D), and δ is the parameter associated with the unobservable confounder Z . The log-likelihood, score, and Hessian contributions of observation i under the full (long) model are

$$\ell_i(\boldsymbol{\theta}) = \log f(Y_i | W_i; \boldsymbol{\theta}), \quad \mathbf{s}_i(\boldsymbol{\theta}) = \frac{\partial \ell_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}, \quad \mathbf{H}_i(\boldsymbol{\theta}) = \frac{\partial^2 \ell_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}. \quad (2.1)$$

Let $H_{i,AA}(\boldsymbol{\theta}) = \partial^2 \ell_i(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}_A \partial \boldsymbol{\theta}_A'$ be the $(p+1) \times (p+1)$ sub-matrix corresponding to the nuisance parameters, and define the sub-blocks $H_{i,DZ}(\boldsymbol{\theta})$, $H_{i,DD}(\boldsymbol{\theta})$, $H_{i,DA}(\boldsymbol{\theta})$, $H_{i,AA}(\boldsymbol{\theta})$, and $H_{i,AZ}(\boldsymbol{\theta})$ of $\mathbf{H}_i(\boldsymbol{\theta})$ accordingly. The population score and information matrix are

$$\mathbf{s}(\boldsymbol{\theta}) := \mathbb{E}[\mathbf{s}_i(\boldsymbol{\theta})], \quad \mathcal{I}(\boldsymbol{\theta}) := -\mathbb{E}[\mathbf{H}_i(\boldsymbol{\theta})], \quad (2.2)$$

where the expectation is taken under the true data-generating process. Because Z_i is unobservable, the researcher fits the short model that conditions only on the observed regressors W_i^* , omitting Z_i . Let $f^*(y | w^*; \boldsymbol{\theta})$ denote the short-model density. The

short-model log likelihood and score contributions are:

$$\ell_i^*(\boldsymbol{\theta}) = \log f^*(Y_i | W_i^*; \boldsymbol{\theta}), \quad \mathbf{s}_i^*(\boldsymbol{\theta}) = \frac{\partial \ell_i^*(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}. \quad (2.3)$$

The short-model maximum likelihood estimator $\hat{\boldsymbol{\theta}}^*$ solves $n^{-1} \sum_i \mathbf{s}_i^*(\hat{\boldsymbol{\theta}}^*) = \mathbf{0}$ and (under regularity conditions) converges in probability³ to the pseudo-true parameter $\boldsymbol{\theta}^*$, defined as (White, 1982):

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} \mathbb{E}[\log f^*(Y_i | W_i^*; \boldsymbol{\theta})]. \quad (2.4)$$

Because the short model omits a relevant variable, the pseudo-true and true parameter values do not coincide, in general. The magnitude of the bias is established in the following result (the proof and specific underpinning assumptions are given in the Appendices A and B).

Lemma 2.1. *Under regularity conditions R1–R6 (detailed in Appendix A), the pseudo-true parameter satisfies:*

$$\beta^* - \beta_0 = \tilde{\pi} \cdot \delta_0 + O(\delta_0^2), \quad (2.5)$$

where

$$\tilde{\pi} = \frac{\tilde{\mathcal{I}}_{DZ}^*}{\tilde{\mathcal{I}}_{DD}^*} = \frac{-\mathbb{E}[\mathbf{H}_{i,DZ}(\boldsymbol{\theta}^*)] + \mathbb{E}[\mathbf{H}_{i,DA}(\boldsymbol{\theta}^*)] \mathbb{E}[\mathbf{H}_{i,AA}(\boldsymbol{\theta}^*)]^{-1} \mathbb{E}[\mathbf{H}_{i,AZ}(\boldsymbol{\theta}^*)]}{-\mathbb{E}[\mathbf{H}_{i,DD}(\boldsymbol{\theta}^*)] + \mathbb{E}[\mathbf{H}_{i,DA}(\boldsymbol{\theta}^*)] \mathbb{E}[\mathbf{H}_{i,AA}(\boldsymbol{\theta}^*)]^{-1} \mathbb{E}[\mathbf{H}_{i,AD}(\boldsymbol{\theta}^*)]}, \quad (2.6)$$

is the ratio of Schur complement entries of the information matrix $\mathcal{I}(\boldsymbol{\theta}^*)$, evaluated at the pseudo-true parameter, after projecting out the nuisance block $\mathcal{I}_{AA}^*$. In particular, since $\hat{\beta}^* \xrightarrow{p} \beta^*$ under the regularity conditions, $\text{plim}_{n \rightarrow \infty}(\hat{\beta}^* - \beta_0) = \tilde{\pi} \cdot \delta_0 + O(\delta_0^2)$.

The quantity $\tilde{\pi}$ measures the extent to which the outcome varies with the omitted confounder Z_i through the interaction with D_i . That is, it captures how much of the variation in the outcome that is currently attributed to D_i by the short model would instead be explained by Z_i if the latter were included. A large $|\tilde{\pi}|$ means that the confounder and the treatment pull the outcome in similar directions after conditioning on the nuisance covariates $\mathbf{X}_i$, so that omitting Z_i causes the short-model estimate β^* to absorb

³The regularity conditions are detailed in Appendix A.

a substantial portion of Z_i 's influence. A value of $\tilde{\pi}$ close to zero means the confounder's effect on the outcome is orthogonal to the treatment's effect after partialling out $\mathbf{X}_i$, so omission of Z_i causes negligible bias regardless of how strongly Z_i affects the outcome in isolation.

Appendix B shows that the parameter $\tilde{\pi}$ also admits an approximate slope interpretation. More specifically, if $\mathcal{I}(\boldsymbol{\theta})$ is continuously differentiable in $\boldsymbol{\theta}$ on Θ , a first-order Taylor expansion around $\boldsymbol{\theta}_0$ gives $\tilde{\pi} = \tilde{\mathcal{I}}_{DZ}^0 / \tilde{\mathcal{I}}_{DD}^0 + O(\delta_0)$, where the superscript 0 denotes evaluation at $\boldsymbol{\theta}_0$. Therefore $\tilde{\pi}$ is approximately (for small confounding) the population slope from regressing the omitted-variable score residual $\tilde{s}_{\delta,i}^0$ on the treatment score residual $\tilde{s}_{\beta,i}^0$, where both residuals are obtained by projecting out the nuisance score vector $\mathbf{s}_{A,i}^0 = \partial \ell_i(\boldsymbol{\theta}_0, 0) / \partial \boldsymbol{\theta}_A$. Therefore, these residuals are the components of the treatment and omitted-variable scores that are orthogonal to the nuisance score vector $\mathbf{s}_{A,i}^0$. This is equivalent to the way in which in linear models the FWL residual is the component of the treatment variable that is orthogonal to the control variables in a linear regression.

For GLMs the structure of the Hessian makes the above interpretation clearer. As shown in Appendix B, for GLMs with variance weight $w_i^* = V(\mu_i(\boldsymbol{\theta}^*))$ evaluated at the pseudo-true parameter,

$$\tilde{\mathcal{I}}_{DD}^* = \mathbb{E}[w_i^* \tilde{D}_i^{*2}], \quad \tilde{\mathcal{I}}_{DZ}^* = \mathbb{E}[w_i^* \tilde{D}_i^* \tilde{Z}_i^*], \quad (2.7)$$

where

$$\begin{aligned} \tilde{D}_i^* &= D_i - (\boldsymbol{\pi}_D^*)' A_i, & \boldsymbol{\pi}_D^* &= \mathbb{E}[w_i^* A_i A_i']^{-1} \mathbb{E}[w_i^* A_i D_i], \\ \tilde{Z}_i^* &= Z_i - (\boldsymbol{\pi}_Z^*)' A_i, & \boldsymbol{\pi}_Z^* &= \mathbb{E}[w_i^* A_i A_i']^{-1} \mathbb{E}[w_i^* A_i Z_i], \end{aligned}$$

are residuals from w^* -weighted projections of D_i and Z_i onto $A_i = (1, \mathbf{X}_i')' \in \mathbb{R}^{p+1}$, and so

$$\tilde{\pi} = \frac{\mathbb{E}[w_i^* \tilde{D}_i^* \tilde{Z}_i^*]}{\mathbb{E}[w_i^* \tilde{D}_i^{*2}]}. \quad (2.8)$$

This is a second-moment ratio and under the same additional differentiability assumption,

$w_i^* = w_i^0 + O(\delta_0)$ where $w_i^0 = V(\mu_i(\boldsymbol{\theta}_0))$, and (2.8) becomes

$$\tilde{\pi} = \frac{\text{Cov}_{w^0}(\tilde{D}_i^0, \tilde{Z}_i^0)}{\text{Var}_{w^0}(\tilde{D}_i^0)} + O(\delta_0), \quad (2.9)$$

where $\tilde{D}_i^0$ and $\tilde{Z}_i^0$ are the residuals from w^0 -weighted projections of D_i and Z_i onto A_i , and Cov_{w^0} , Var_{w^0} denote w^0 -weighted covariance and variance. Hence, in a neighbourhood of $\delta_0 = 0$, $\tilde{\pi}$ is approximately the population regression slope from regressing $\tilde{Z}_i^0$ on $\tilde{D}_i^0$ with weights w_i^0 .

Equation (2.9) is a nonlinear equivalent of the OLS treatment residual slope in Cinelli and Hazlett (2020). Specifically, they show that given the long linear regression model $Y_i = \beta_0 D_i + \boldsymbol{\gamma}'_0 \mathbf{X}_i + \delta_0 Z_i + \varepsilon_i$, with $\mathbb{E}[\varepsilon_i \mid D_i, \mathbf{X}_i, Z_i] = 0$, and the corresponding short model $Y_i = \beta^* D_i + \boldsymbol{\gamma}^* \mathbf{X}_i + \varepsilon_i^*$, the Frisch-Waugh-Lovell theorem at the population level implies that

$$\beta^* - \beta_0 = \delta_0 \underbrace{\frac{\text{Cov}(\tilde{Z}_i, \tilde{D}_i)}{\text{Var}(\tilde{D}_i)}}_{\tilde{\pi}} \quad (2.10)$$

where $\tilde{D}_i$ and $\tilde{Z}_i$ are the residuals from OLS projections of D and Z onto $A = (1, \mathbf{X}')$, which correspond to the unit-weighted ($w_i \equiv 1$) special case of the w -weighted projections in the nonlinear expression above

The structural similarity with (2.10) might suggest that the Cinelli–Hazlett methodology could be applied to likelihood-based models using a suitably defined residual. For example, for a binary logit one could use the standardised residuals from the short model:

$$\tilde{Y}_i = \frac{Y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1 - \hat{\mu}_i)}}, \quad \tilde{D}_i = D_i - \hat{\pi}_i, \quad (2.11)$$

where $\hat{\mu}_i = \Lambda(\hat{\alpha}^* + \hat{\beta}^* D_i + \hat{\boldsymbol{\gamma}}^* \mathbf{X}_i)$ is the fitted probability from the short logit model and $\hat{\pi}_i$ is the fitted value from a linear regression of D_i on $\mathbf{X}_i$. One could then compute the partial correlation between these residuals and construct partial R^2 analogues:

$$R_{Y \sim Z \mid \mathbf{X}, D}^2 \approx \text{Corr}^2(\tilde{Y}_i, \tilde{Z}_i), \quad R_{D \sim Z \mid \mathbf{X}}^2 \approx \text{Corr}^2(\tilde{D}_i, \tilde{Z}_i), \quad (2.12)$$

and formally apply the Cinelli–Hazlett bound as if the model were linear.

By Lemma 2.1, however, the procedure just described is acceptable only when the amount of confounding is very small. The nonlinear OVB formula (2.5) contains a non-trivial remainder R of order $O(\delta_0^2)$. This implies that stronger confounding produces a larger remainder and a more severe approximation error. Further, as shown in the Appendix B, the direction and first-order magnitude of $\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*$, which governs R through the $O(\delta_0^2)$ term, is determined by the quantity

$$\text{Cov}[\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)], \quad (2.13)$$

where $\mathbf{s}_i^*(\boldsymbol{\theta}_0)$ is the short-model score at the truth and $s_{i,\delta}(\boldsymbol{\theta}_0, 0) = \partial \ell_i / \partial \delta|_{\delta_0=0}$ is the score of the true model with respect to the omitted coefficient evaluated at the correctly specified parameter. This covariance is zero if and only if Z_i carries no information about the short-model residuals at the truth. In that case omitting Z_i causes no first order distortion of the pseudo-true parameter⁴.

3 Synthetic Confounder Sensitivity Analysis

The key element in the Cinelli and Hazlett (2020) approach are two partial R^2 quantities selected by the researcher: $R_{D \sim Z|\mathbf{X}}^2$, the share of residual treatment variance explained by the confounder, and $R_{Y \sim Z|D,\mathbf{X}}^2$, the share of residual outcome variance explained by the confounder after conditioning on treatment. Because the linear OVB formula is exact, closed-form bounds on the bias can be expressed directly as functions of these two quantities, and their worst-case values over the feasibility region yield sharp, closed form sensitivity intervals.

The remainder in the nonlinear OVB formula prevents exact closed-form character-

⁴The remainder R is zero in the linear case. Linear models solve a problem of the form $\ell_i(\boldsymbol{\theta}) = -\frac{1}{2\sigma^2}(Y_i - W_i'\boldsymbol{\theta})^2 + \text{const}$, which is quadratic in $\boldsymbol{\theta}$. Its Hessian $\mathbf{H}_i(\boldsymbol{\theta}) = -W_i W_i' / \sigma^2$ is therefore constant (as it does not depend on $\boldsymbol{\theta}$). As a result, the Taylor expansion of the population score around $\boldsymbol{\theta}^*$ is exact at second order, and the implicit-function expansion of $\boldsymbol{\theta}^*(\delta_0)$ around $\delta_0 = 0$ (Lemma A.1 in the Appendix) terminates exactly at first order. The remainder R is therefore identically zero, and the main results in Cinelli and Hazlett (2020) are recovered as a special case of Lemma 2.1.

isation of the bias. However closed-form bounds are not essential to develop a viable sensitivity framework for nonlinear models. Rather this paper constructs a synthetic confounder with researcher-selected partial correlations. This synthetic confounder is included into the short model, and the model’s marginal effects are then re-estimated, thus approximating the effect of a confounder with specific partial correlations. In this approach, the sensitivity parameters $\rho_D = \text{Cor}(Z, \tilde{D})$ and $\rho_Y = \text{Cor}(Z, \tilde{Y})$ play the same role as $\sqrt{R_{D \sim Z|\mathbf{X}}^2}$ and $\sqrt{R_{Y \sim Z|D, \mathbf{X}}^2}$ in Cinelli and Hazlett (2020). As in Cinelli and Hazlett (2020), the simulation approach presented here delivers “exact” adjusted AME at each (ρ_D, ρ_Y) point rather than a first-order approximation to it.

Specifically, suppose

$$\mathbb{E}[Y_i | D_i, \mathbf{X}_i, Z_i] = h(\alpha + \beta D_i + \boldsymbol{\gamma}'\mathbf{X}_i + \delta Z_i), \quad (3.1)$$

where $h : \mathbb{R} \rightarrow \mathcal{Y}$ is a known, invertible mean function. Note that this expression does not include an interaction between Z and D for convenience, and the extension to relax this homogeneity condition is straightforward, as is discussed below. Writing $g = h^{-1}$ for the link function, equation (3.1) is equivalent to

$$\eta_i = g(\mu_i) = \alpha + \beta D_i + \boldsymbol{\gamma}'\mathbf{X}_i + \delta Z_i, \quad \mu_i = \mathbb{E}[Y_i | D_i, \mathbf{X}_i, Z_i]. \quad (3.2)$$

This single-index structure encompasses all GLMs (logit, probit, Poisson, negative binomial), Tobit, ordered probit, and the fractional response model of Papke and Wooldridge (1996), among others. The parameter of interest is the average marginal effect (AME):

$$\overline{\text{AME}} = \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial D_i} \mathbb{E}[Y_i | D_i, \mathbf{X}_i, Z_i] = \frac{1}{n} \sum_{i=1}^n \beta h'(\eta_i). \quad (3.3)$$

Because Z_i is unobservable, the researcher estimates the short model

$$\eta_i^* = g(\mu_i^*) = \alpha^* + \beta^* D_i + \boldsymbol{\gamma}^*\mathbf{X}_i. \quad (3.4)$$

Let $\hat{\beta}^*$ denote the short-model MLE, and let the short-model AME be $\widehat{\overline{\text{AME}}}^* =$

$n^{-1} \sum_i \hat{\beta}^* h'(\hat{\eta}_i^*)$. The objective of the sensitivity analysis is to characterise how $\widehat{\text{AME}}^*$ changes under the influence of confounders of different strengths.

As in Cinelli and Hazlett (2020), our method relies on the residualised treatment, $\tilde{D}_i$ (that is the component of D_i orthogonal to $\mathbf{X}_i$) and the residualised outcome $\tilde{Y}_i$ (a measure of outcome variation unexplained by D_i and $\mathbf{X}_i$) under the short model. These are assumed to be available (the construction of the residuals is detailed in the following subsection). Once these quantities are available, the researcher fixes the levels of the partial correlations

$$\rho_D = \text{Cor}(Z, \tilde{D}), \quad \rho_Y = \text{Cor}(Z, \tilde{Y}). \quad (3.5)$$

The parameters ρ_D and ρ_Y are the nonlinear-model equivalents of the partial R^2 quantities $R_{D \sim Z|\mathbf{X}}^2$ and $R_{Y \sim Z|D, \mathbf{X}}^2$ used by Cinelli and Hazlett (2020) for the linear model. When set by the researcher, (ρ_D, ρ_Y) will normally take on non-zero values, reflecting the researcher's beliefs about the amount of confounding.

For given ρ_D and ρ_Y , we can now construct a synthetic confounder. Let $\tilde{D}^*$ and $\tilde{Y}^*$ denote the standardised residuals (zero mean, unit variance), and let $r = \text{Cor}(\tilde{D}^*, \tilde{Y}^*)$ be their sample correlation. Then, we have the following result:

Proposition 3.1. *The variable*

$$Z = a \tilde{D}^* + b \tilde{Y}^* + c \varepsilon, \text{ where } \varepsilon \sim i.i.d.N(0, 1) \quad (3.6)$$

satisfies $\text{Cor}(Z, \tilde{D}^) = \rho_D$ and $\text{Cor}(Z, \tilde{Y}^*) = \rho_Y$ when*

$$a = \frac{\rho_D - \rho_Y r}{1 - r^2}, \quad b = \frac{\rho_Y - \rho_D r}{1 - r^2}, \quad c = \sqrt{1 - a^2 - b^2 - 2abr}. \quad (3.7)$$

The construction is feasible if and only if $c^2 \geq 0$, which requires

$$\rho_D^2 + \rho_Y^2 - 2\rho_D \rho_Y r \leq 1 - r^2. \quad (3.8)$$

The proof is in Appendix C. The quantities a and b are the solution to a system of two

linear equations with given ρ_D and ρ_Y . The quantity c is determined by the normalisation $\text{Var}(Z) = 1$. The feasibility condition (3.8) determines the set of achievable (ρ_D, ρ_Y) pairs, and it reduces to the unit circle when $r = 0$. The quantity r is a nuisance quantity that guarantees that $\text{Cor}(Z, \tilde{D}^*) = \rho_D$ and $\text{Cor}(Z, \tilde{Y}^*) = \rho_Y$ exactly ⁵.

Given Proposition 3.1, the synthetic confounder sensitivity analysis is as follows. For a user-specified grid of (ρ_D, ρ_Y) pairs:

1. Compute a, b, c from (3.7); check feasibility via (3.8).
2. Draw $\varepsilon \sim N(0, 1)$.
3. Construct Z via (3.6) and augment the short model with Z as an additional regressor.
4. Re-estimate the long model (3.2) by maximum likelihood and compute the adjusted AME (3.3).

The above procedure allows us map each feasible (ρ_D, ρ_Y) to an adjusted AME. If the mapping excludes zero for a given ρ_D , the estimated effect is judged robust to confounding of that strength. Because Z is constructed and the full model is re-estimated numerically, no Taylor approximation is made and the remainder R in the OVB formula (2.5) is implicitly absorbed without requiring it to be small.

3.1 Construction of residuals.

The discussion so far assumed the existence of suitable standardized residuals. In practice, these need to be computed as a first step in such a way that they capture variation

⁵The r is 0 or near zero for a large family of models. For example, for GLM, letting $e_i = Y_i - \hat{\mu}_i$ be the raw prediction error from the short model, and $\tilde{D}_i$ be the OLS treatment residual. The MLE satisfies the normal equations $\sum_i (Y_i - \hat{\mu}_i) D_i = 0$ and $\sum_i (Y_i - \hat{\mu}_i) X_{ij} = 0$ for all j . Since $\tilde{D}_i = D_i - \sum_j X_{ij} \hat{\beta}_j$ is a linear combination of D_i and $\{X_{ij}\}_{j=1}^p$, then

$$\sum_i e_i \tilde{D}_i = \sum_i (Y_i - \hat{\mu}_i) D_i - \sum_j \hat{\beta}_j \sum_i (Y_i - \hat{\mu}_i) X_{ij} = 0$$

In practice we rely on standardised residuals. Dividing by $\sqrt{\hat{\mu}_i(1 - \hat{\mu}_i)}$ introduces observation-specific weights that break exact orthogonality, so $r = \text{Cor}(\tilde{D}^*, \tilde{Y}^*)$ is approximately but not exactly zero (of order 10^{-4} in well-specified models). For this reason, in practice we compute r from the data rather than fixing it to zero. This is particularly to improve the precision of the bisection method described in later sections.

unexplained by $\mathbf{X}$ (and D) in the short model. The natural choice in each model family is the standardised residual

$$\tilde{Y}_i = \frac{Y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}, \quad (3.9)$$

where $\hat{\mu}_i$ is the short-model fitted value and $V(\cdot)$ is the variance function of the model family. We give three examples next.

Example 3.1. Consider first a binary logit, which gives fitted probabilities $\hat{\mu}_i = \Lambda(\hat{\eta}_i^*)$ where Λ is the logistic CDF and $\hat{\eta}_i^* = \hat{\alpha}^* + \hat{\beta}^* D_i + \hat{\gamma}^{*\prime} \mathbf{X}_i$. The variance function is $V(\mu) = \mu(1 - \mu)$, so the standardised residual is

$$\tilde{Y}_i = \frac{Y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1 - \hat{\mu}_i)}}. \quad (3.10)$$

which is already scaled by the conditional standard deviation of Y_i . This residual is then used in Proposition 3.1.

Example 3.2. Consider, next, the standard Tobit model with latent outcome $Y_i^* = \alpha + \beta D_i + \gamma' \mathbf{X}_i + u_i$, $u_i \sim N(0, \sigma^2)$, and observed outcome $Y_i = \max(0, Y_i^*)$. The conditional mean and variance of the observed censored outcome are

$$\hat{E}[Y_i | \mathbf{X}_i] = \Phi(z_i)(\mathbf{X}_i' \hat{\boldsymbol{\beta}} + \hat{\sigma} \hat{\lambda}_i), \quad (3.11)$$

$$\widehat{Var}[Y_i | \mathbf{X}_i] = \hat{\sigma}^2 \Phi(z_i)[1 - \Phi(z_i)(\hat{z}_i + \hat{\lambda}_i)^2], \quad (3.12)$$

where $z_i = \mathbf{X}_i' \hat{\boldsymbol{\beta}} / \hat{\sigma}$ and $\hat{\lambda}_i = \phi(z_i) / \Phi(-z_i)$ is the inverse Mills ratio. The standardised residual for Tobit is therefore

$$\tilde{Y}_i = \frac{Y_i - \hat{E}[Y_i | \mathbf{X}_i]}{\sqrt{\widehat{Var}[Y_i | \mathbf{X}_i]}}. \quad (3.13)$$

3.2 Treatment Residuals

Our method also requires the treatment residuals. We can use the standard OLS residual $\hat{D}_i = D_i - \mathbf{X}_i' \hat{b}^{\text{OLS}}$, $\hat{b}^{\text{OLS}} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' D$, which is then standardised to zero mean and

unit variance to obtain

$$\tilde{D}_i^* = \frac{\hat{D}_i - \bar{\hat{D}}}{\hat{\sigma}_{\hat{D}}},$$

where $\bar{\hat{D}}$ and $\hat{\sigma}_{\hat{D}}$ are the sample mean and standard deviation of $\hat{D}_i$. It is $\tilde{D}_i^*$ that enters Proposition 3.1 as the treatment residual primitive. When D is binary one might consider replacing the OLS residual $\hat{D}_i$ with a probit or logit residual $D_i - \Phi(\mathbf{X}_i' \hat{\gamma})$ before standardising. However, the construction of Z in Proposition 3.1 is valid for any choice of treatment residual prior to standardisation. Specifically, the loadings a, b, c adjust automatically through (3.7), so that the target correlations ρ_D and ρ_Y are achieved exactly regardless of which residual is standardised. The choice only affects the interpretation of ρ_D . In practice we use the OLS residual throughout for two reasons. First, ρ_D then matches the Cinelli–Hazlett partial correlation exactly via the linear FWL identity, facilitating benchmarking against that framework. Second, for moderate propensity scores the OLS and probit residuals are nearly proportional, so the choice has little practical impact.

Note that, by construction, $\tilde{D}^*$ and $\tilde{Y}^*$ are scalar, standardised residuals. All of the nonlinearity in the original model due to the curvature of $h(\cdot)$ or the discreteness of Y_i itself has already been absorbed into the construction of $\tilde{Y}^*$ by the time $\rho_Y = \text{Cor}(Z, \tilde{Y}^*)$ is computed. Correlating Z against $\tilde{Y}^*$ is therefore just a linear operation between two scalars. This means that the relationship $R^2 = \rho^2$ applies here and allows a specific interpretation of the results. Specifically there is a hypothetical regression of $\tilde{Y}^*$ on Z with $R_{\tilde{Y}^* \sim Z}^2 = \rho_Y^2$ exactly. This R^2 measures the share of variance in the standardised residual explained by Z .

In practice, this means that a researcher selecting $\rho_Y = 0.3$, for instance, is equivalently specifying a hypothetical confounder that explains $\rho_Y^2 = 9\%$ of the variance in the outcome residual (the part of Y_i left over after D_i and $\mathbf{X}_i$ have already been accounted for), and similarly for $\tilde{D}^*$ (a chosen $\rho_D = 0.2$ corresponds to a confounder explaining 4% of the residual variation in D_i once $\mathbf{X}_i$ has been partialled out).

3.3 Homogeneity

The single-index structure (3.1) assumes that the confounder Z_i enters the outcome equation additively with a homogeneous coefficient δ , ruling out interactions between Z_i and the treatment D_i . This assumption is made for expositional convenience and is not required. To see why, note that Z is built entirely from the residuals $\tilde{D}^*$ and $\tilde{Y}^*$ of the short model via Proposition 3.1, without any reference to a structural coefficient on Z or to how Z enters the outcome equation. The researcher specifies only the partial correlations (ρ_D, ρ_Y) of the synthetic confounder with the residualised treatment and outcome and this determines (a, b, c) (which carry no information about whether the true confounding mechanism involves an interaction with D).

Once the synthetic confounder Z has been constructed, the researcher can explicitly accommodate treatment-effect heterogeneity by augmenting the re-estimated model with the interaction term $Z_i \times D_i$:

$$g(\mu_i) = \alpha + \beta D_i + \gamma' \mathbf{X}_i + \delta Z_i + \phi \{Z_i \times D_i\}, \quad (3.14)$$

and re-estimating (3.14) by maximum likelihood in place of (3.2). The adjusted AME computed from (3.14) then reflects the sensitivity of the treatment effect to a confounder whose influence on the outcome varies with the level of D_i .

4 Fragility Thresholds and Benchmarking

The method described in the previous section allows researchers to fix the specific values of the partial correlations, enabling the characterisation of a sensitivity surface. In practice, however, one desires to know how strong the confounding must be to overturn a result. More specifically, as noted in Cinelli and Hazlett (2020), Masten and Poirier (2022) and Diegert et al. (2025), there are two natural thresholds to consider, namely the explain-away threshold and the sign change threshold.

Consider a fixed level of ρ_Y and define $g(\rho_D) = \widehat{\text{AME}}(\rho_D, \rho_Y)$. The feasibility condi-

tion (3.8) defines a feasible range of variation for ρ_D . Setting $c^2 = 0$ and solving for ρ_D , we obtain two roots,

$$\rho_D^{\pm \text{bdy}} = r\rho_Y \pm \sqrt{(1-r^2)(1-\rho_Y^2)}. \quad (4.1)$$

Within the feasible interval defined by these roots, $[\rho_D^{-\text{bdy}}, \rho_D^{+\text{bdy}}]$, we can first define the explain-away threshold

$$\rho_D^*(\rho_Y) = \arg \min_{|\rho_D|} \left\{ \rho_D \in [\rho_D^{-\text{bdy}}, \rho_D^{+\text{bdy}}] : |g(\rho_D) / \widehat{\text{SE}}(\rho_D)| < z_{\alpha/2} \right\} \quad (4.2)$$

where $\widehat{\text{SE}}(\rho_D)$ is the standard error of the AME. In other words, $\rho_D^*(\rho_Y)$ constitutes the minimal amount of confounding required to overturn the statistical significance of AME. Similarly, the sign-change threshold

$$\rho_D^+(\rho_Y) = \arg \min_{|\rho_D|} \left\{ \rho_D \in [\rho_D^{-\text{bdy}}, \rho_D^{+\text{bdy}}] : \text{sign}(g(\rho_D)) \neq \text{sign}(g(0)) \right\} \quad (4.3)$$

is the feasible value closest to zero at which the direction of the estimated effect reverses.

Computing ρ_D^* and ρ_D^+ requires searching over the feasible interval. We therefore rely on a grid-search followed by a bisection method to compute the values⁶. More specifically, in a first stage, the synthetic method is applied on a grid of equally-spaced points within $[\rho_D^{-\text{bdy}}, \rho_D^{+\text{bdy}}]$. This search identifies an initial pair of grid points where the criterion switches from unmet to met. Within that interval we can apply a standard bisection algorithm to find the root of the function. The two thresholds ρ_D^* and ρ_D^+ are computed independently by the same algorithm with different criterion functions: $|t(\rho_D)| < z_{\alpha/2}$ for explain-away and $\text{sign}(g(\rho_D)) \neq \text{sign}(g(0))$ for sign-change. If no grid point triggers the criterion, the result is reported as robust for that threshold⁷.

⁶Instead of jsut using just a bisection method. In a linear model the bias is proportional to ρ_D , so the AME is monotone in ρ_D for fixed ρ_Y . In a nonlinear, in principle, for fixed ρ_Y , the greater ρ_D , the greater the weight Z plays and the smallest the coefficient of the variable D . That would suggest that the change-sign criterion changes monotonically. However, the explain away criteria depends on the the AME, as well as the standard errors, and both criteria depend on the curvature of the nonlinear likelihood model, so monotonicity might not be guaranteed, although the results in the simulations and the application did not yield any finding to suggest otherwise

⁷It is convenient to emphasise here that in the implementation of the algorithm ε in (3.6) is fixed across all grid evaluations. This ensures that $g(\rho_D)$ varies smoothly as a function of ρ_D , which makes both the grid search reliable and the bisection refinement well-behaved.

4.1 Benchmarking

As described so far, the synthetic confounder approach (as the Cinelli and Hazlett, 2020 framework) is subject to two main limitations. First, the values of ρ_Y (and ρ_D) are pre-specified by the researcher and thus subject to some subjectivity (and potentially detached from the specific application at hand). Second, and more importantly, the fragility thresholds ρ_D^* and ρ_D^+ measure the confounding strength required to render a result statistically insignificant or to reverse its sign, yet a heavily biased result can appear robust precisely because the bias inflates the initial t -statistic⁸.

The solution proposed by Cinelli and Hazlett (2020) for both problems is benchmarking. Rather than asking whether any hypothetical confounder can explain away the result, benchmarking asks how much the estimate would shift if a confounder as strong as a specific observed variable had been omitted, grounding the choice of (ρ_D, ρ_Y) in the data and resolving the subjectivity problem. This also addresses the inflated-baseline problem. If the researcher can identify an observed covariate whose partial correlations with treatment and outcome are at least as large as those of the true omitted confounder, then the benchmark-implied correction will be large enough to reveal that the short-model estimate is substantially driven by confounding, regardless of how inflated the initial t -statistic is. Note, however, that the credibility of any ensuing conclusion depends on the choice of benchmark. A covariate that is strongly correlated with the treatment after conditioning on the remaining controls thus presents a potential upper bound on what a similar unobserved confounder could do.

As in the Cinelli–Hazlett framework, the benchmarking procedure for likelihood based models computes $(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k})$ for each candidate variable W_k , and then it evaluates the adjusted AME at those levels. Comparing $\hat{\rho}_{D,k}$ to $\rho_D^*(\hat{\rho}_{Y,k})$ one can explore whether a confounder of W_k ’s strength would explain away the result, while the adjusted AME and implied bias quantify how much the short-model estimate would be overstated.

⁸To see this last point, note that the explain-away threshold ρ_D^* tells us how large must ρ_D be to drive $|t(\rho_D)| < z_{\alpha/2}$. Since $t(\rho_D) = \widehat{\text{AME}}(\rho_D) / \widehat{\text{SE}}(\rho_D)$, a large threshold means that the adjusted AME must fall a long way before it becomes insignificant. But if the short model AME is itself inflated by omitted variable bias, the starting point is artificially high, and the distance to insignificance is correspondingly large. The threshold is therefore measuring distance from a biased baseline, not distance from the truth.

More precisely, the procedure is implemented as follows. Let $\mathbf{X}_{-k}$ denote the control vector with W_k removed. Then:

1. Regress W_k on $\mathbf{X}_{-k}$ by OLS and save the standardised residuals

$$W_{r,k} = \frac{W_k - \mathbf{X}_{-k}\hat{b}}{\text{SD}(W_k - \mathbf{X}_{-k}\hat{b})}.$$

This partialling-out removes variation in W_k that is collinear with the remaining controls, so that the subsequent correlations measure the partial association of W_k with treatment and outcome after conditioning on everything else in $\mathbf{X}$.

2. Re-estimate the OLS regression of D on $\mathbf{X}_{-k}$ (excluding W_k) and save the standardised residuals $\tilde{D}_{-k}$. Compute

$$\hat{\rho}_{D,k} = \text{Cor}(W_{r,k}, \tilde{D}_{-k}).$$

3. Re-estimate the same nonlinear model used for the main analysis, $\text{model}(Y) \sim D + \mathbf{X}_{-k}$, and compute standardised standardised residuals $\tilde{Y}_{-k}$ from the fitted values of this reduced model. For binary, ordered, and count models these are

$$\tilde{Y}_{-k,i} = \frac{Y_i - \hat{\mu}_{-k,i}}{v(\hat{\mu}_{-k,i})^{1/2}},$$

where $v(\cdot)$ is the variance function of the model family (e.g. $\hat{\mu}(1 - \hat{\mu})$ for logit and probit, $\hat{\mu}$ for Poisson, $\hat{\mu} + \hat{\phi}\hat{\mu}^2$ for negative binomial). Compute

$$\hat{\rho}_{Y,k} = \text{Cor}(W_{r,k}, \tilde{Y}_{-k}),$$

or $\hat{\rho}_{Y,k,j} = \text{Cor}(W_{r,k}, \tilde{Y}_{-k,j})$ for each outcome j in the mlogit case.

4. Check feasibility of $(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k})$ via the loading formulas (3.7). If feasible, construct Z using the main-model residuals $\tilde{D}$ and $\tilde{Y}$, fit the long model, and record the

adjusted AME, SE, t -statistic, and

$$\text{bias} = \widehat{\text{AME}}_{\text{short}} - \widehat{\text{AME}}(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k}).$$

A large bias indicates that omitting a confounder as influential as W_k would materially distort the estimated treatment effect.

The benchmark output reports, for each variable W_k , $\hat{\rho}_{D,k}$ and $\hat{\rho}_{Y,k}$ can be compared to the bisection thresholds. If $\hat{\rho}_{D,k} < \rho_D^*(\hat{\rho}_{Y,k})$, the benchmarking variable's confounding strength falls short of the explain-away threshold. We would then conclude that even if such a variable were omitted, the result would remain statistically significant. Conversely, if $\hat{\rho}_{D,k} > \rho_D^*(\hat{\rho}_{Y,k})$, the omission of W_k would have been enough to explain away the result.

Example 4.1. *In a multinomial logit with $J > 2$ categories, each category has its own set of structural parameters and its own residual. In particular, with J categories and baseline category $j = 0$, we have*

$$P(Y_i = j \mid D_i, \mathbf{X}_i, Z_i) = \frac{\exp(\alpha_j + \beta_j D_i + \boldsymbol{\gamma}'_j \mathbf{X}_i + \delta_j Z_i)}{1 + \sum_{k \neq 0} \exp(\alpha_k + \beta_k D_i + \boldsymbol{\gamma}'_k \mathbf{X}_i + \delta_k Z_i)}, \quad j \neq 0. \quad (4.4)$$

Each category j has its own coefficient on the omitted confounder δ_j . The average marginal effect of D on $P(Y_i = j)$ is

$$\widehat{\text{AME}}_j = \frac{1}{n} \sum_{i=1}^n \hat{P}(Y_i = j) [\beta_j - \sum_{k \neq 0} \beta_k \hat{P}(Y_i = k)], \quad (4.5)$$

and each category produces a separate AME that must be assessed independently.

Define the residual for category j as

$$\tilde{Y}_{j,i} = \mathbf{1}(Y_i = j) - \hat{P}(Y_i = j \mid D_i, \mathbf{X}_i), \quad j = 1, \dots, J. \quad (4.6)$$

If one computes Z using only category j 's residual, then Z achieves the target ρ_Y with $\tilde{Y}_j$ by construction, but not with the remaining $\tilde{Y}_k$ for $k \neq j$. The consequence is that including such a Z in the multinomial logit would simultaneously shift the AMEs of all J

categories, but in uncontrolled directions for $k \neq j$, and we cannot interpret the resulting sensitivity surface for category j as the effect of a confounder with a specific (ρ_D, ρ_Y) relationship to category j alone.

The solution proposed here is to construct a separate synthetic confounder Z_j for each outcome category j . Each Z_j is built using category j 's own residual $\tilde{Y}_j$ in the loading formula, so that its target correlations (ρ_D, ρ_Y) are defined with respect to outcome j . The multinomial logit is then re-estimated with Z_j included and the AME for category j is recorded. This is repeated independently for every category.

Formally, for each category j :

1. Compute the standardised residuals from the short model: $\tilde{Y}_{j,i} = \mathbf{1}(Y_i = j) - \hat{P}(Y_i = j | D_i, \mathbf{X}_i)$, and standardise: $\tilde{Y}_j^* = (\tilde{Y}_j - \bar{\tilde{Y}}_j) / \hat{\sigma}_{\tilde{Y}_j}$.
2. Compute the treatment–outcome residual correlation $r_j = \text{Cor}(\tilde{D}^*, \tilde{Y}_j^*)$.
3. Given target (ρ_D, ρ_Y) and r_j , solve for the loadings (a_j, b_j, c_j) via:

$$a_j = \frac{\rho_D - \rho_Y r_j}{1 - r_j^2}, \quad b_j = \frac{\rho_Y - \rho_D r_j}{1 - r_j^2}, \quad c_j = \sqrt{1 - a_j^2 - b_j^2 - 2a_j b_j r_j}.$$

4. Construct $Z_j = a_j \tilde{D}^* + b_j \tilde{Y}_j^* + c_j \varepsilon$, where $\varepsilon \sim N(0, 1)$ independent of everything else. By construction, $\text{Cor}(Z_j, \tilde{D}^*) = \rho_D$ and $\text{Cor}(Z_j, \tilde{Y}_j^*) = \rho_Y$.
5. Re-estimate the multinomial logit including Z_j as an additional covariate and record $\widehat{AME}_j$.

Steps 1–5 are repeated for each (ρ_D, ρ_Y) combination in the grid and for each category $j \in \{1, \dots, J\}$.

Unlike in Examples 3.1 and 3.2 There will be an outcome-specific feasibility region requiring $c_j^2 \geq 0$, which expands to:

$$1 - a_j^2 - b_j^2 - 2a_j b_j r_j \geq 0. \tag{4.7}$$

with the feasible set of (ρ_D, ρ_Y) being the interior of the ellipse:

$$\rho_D^2 + \rho_Y^2 - 2r_j\rho_D\rho_Y \leq 1 - r_j^2, \quad (4.8)$$

The bisection for the explain-away threshold ρ_D^* and sign-change threshold ρ_D^+ is executed separately for each outcome category using that category's own r_j , residual $\tilde{Y}_j^*$, and confounder Z_j .

Note that the ε draw used to construct Z_j is held fixed across all (ρ_D, ρ_Y) grid points within each category, but is shared across categories within the same grid call. This ensures that the sensitivity surface for each category is smooth (the fixed ε eliminates Monte Carlo noise).

5 Simulation Evidence

This section presents a simulation study aimed at evaluating the performance of the method. The study consists of 3 parts. First we study behaviour under different confounding. Next we verify the correct functioning of the benchmarking procedure and explore its limitations. Finally, we explore whether the theoretical results in Section 2 matter in practice.

5.1 Sensitivity analysis under varying confounding

The first study evaluates the performance of the synthetic confounder under different confounding scenarios. We drew samples of $n = 20,000$ observations from four different data generating process (DGPs). In each DGP, the outcome follows a binary logit model

$$Y \sim \text{logit}(-0.5 + \beta_0 D + 0.3X_1 + 0.2X_2 + \delta_0 Z) \quad (5.1)$$

where $X_j \sim N(0, 1)$, $j = 1, 2$. The treatment is $D = \mathbf{1}(0.3X_1 + u > 0)$, $u \sim N(0, 1)$. The confounder, Z is constructed as $Z = \rho_D^{\text{true}} \tilde{D} + \sqrt{1 - (\rho_D^{\text{true}})^2} \eta$, where $\eta \sim N(0, 1)$, so that $\text{Cor}(Z, \tilde{D}) = \rho_D^{\text{true}}$ by construction. The variables X_j, u and η are i.i.d. We considered

the following scenarios:

- DGP 1, No confounder: $\beta_0 = 0.5, \delta_0 = 0$
- DGP 2, Weak confounder: $\beta_0 = 0.5, \delta_0 = 0.2, \rho_D^{\text{true}} = 0.3 \rightarrow \rho_Y^{\text{true}} \approx 0.09$.
- DGP 3, Strong confounder: $\beta_0 = 0.3, \delta_0 = 0.8, \rho_D^{\text{true}} = 0.6 \rightarrow \rho_Y^{\text{true}} \approx 0.22$.
- DGP 4, Spurious significance: $\beta_0 = 0, \delta_0 = 0.8, \rho_D^{\text{true}} = 0.4 \rightarrow \rho_Y^{\text{true}} \approx 0.29$

Table 1: True AMEs, short-model estimates, and confounding bias by DGP

DGP	β_D	δ_Z	ρ_D^{true}	True AME	Short-model AME	Bias
1 (No confounder)	0.5	0.0	0.5	0.1170	0.1101	−0.0068
2 (Weak confounder)	0.5	0.2	0.3	0.1153	0.1335	0.0183
3 (Strong confounder)	0.3	0.8	0.6	0.0608	0.2650	0.2041
4 (Spurious significance)	0.0	0.8	0.4	0.0000	0.1227	0.1227

Notes: β_D is the logit coefficient on D in the true model; δ_Z is the coefficient on the omitted confounder Z . Short-model AME is estimated on the $n = 500,000$ simulation sample omitting Z . The bias = Short-model AME − True AME.

Table 1 describes the properties of short-model estimates under these scenarios⁹ alongside the short-model estimates. There is no confounding in DGP 1 and, as a result, the short model AME in Table 1 is only subject to sampling noise. In DGP 2, the confounder has a small direct effect on the outcome and moderate association with treatment, producing a modest upward bias of 0.0183 (14%). In DGP 3, the treatment coefficient is small relative to the confounder creating a scenario of severe upward bias. The short-model AME of 0.265 overstates the true AME four-fold. DGP 4 is the most damaging scenario from a policy and research perspective. The true treatment effect is exactly zero, but the strong confounder creates a significant ($t \approx 18$) short-model AME of 0.123 that is entirely spurious.

Table 2 reports the fragility thresholds for all four DGPs. Each element in Table 2 measures, for an omitted confounder that has outcome-residual association ρ_Y , how strongly would the confounder need to be correlated with treatment before the result

⁹For each DGP, the short model AME and the implied ρ_Y^{true} are computed from a simulated sample of $n = 500,000$ draws from the true DGP

Table 2: Fragility thresholds by DGP and ρ_Y .

ρ_Y	DGP 1		DGP 2		DGP 3		DGP 4	
	ρ_D^*	ρ_D^+	ρ_D^*	ρ_D^+	ρ_D^*	ρ_D^+	ρ_D^*	ρ_D^+
-0.5	-0.312	-0.361	-0.411	-0.471	robust	robust	-0.363	-0.415
-0.4	-0.352	-0.414	-0.478	-0.564	robust	robust	-0.410	-0.481
-0.3	-0.450	-0.557	-0.677	robust	robust	robust	-0.534	-0.675
-0.2	robust	robust	robust	robust	robust	robust	robust	robust
-0.1	robust	robust	robust	robust	robust	robust	robust	robust
0.0	0.980	0.997	0.983	0.996	0.993	robust	0.995	robust
0.1	0.583	0.630	0.641	0.678	0.809	0.823	0.637	0.677
0.2	0.396	0.437	0.456	0.492	0.678	0.694	0.443	0.481
0.3	0.295	0.327	0.348	0.377	0.577	0.593	0.331	0.363
0.4	0.234	0.260	0.279	0.303	0.500	0.514	0.263	0.288
0.5	0.196	0.216	0.232	0.252	0.439	0.452	0.218	0.238

ρ_D^* is the explain away threshold (the point at which the adjusted AME becomes statistically insignificant; $|t| < 1.96$). ρ_D^+ is the threshold at which the sign of the adjusted AME reverses. “robust” indicates that no threshold is reached within the feasible range.

is explained away or changes sign. A small threshold indicates that even a weak confounder would suffice. In the table, “robust” indicates that no feasible confounder of that type can overturn the result.

Reading across DGPs at any fixed $\rho_Y > 0$, the thresholds for DGPs 1, 2 and 4 are relatively similar, while the results obtained with DGP 3 requires a strong confounder to be wiped out. As discussed in Section 4.1, the latter reflects the inflated t -statistic in DGP 3 rather than robustness. The confounding bias has made the short-model result look so significant that even a powerful synthetic confounder cannot drive it to insignificance without very high ρ_D . Reading down any column, the thresholds fall monotonically as ρ_Y increases from 0 to 0.5, because a confounder more strongly associated with the outcome needs a smaller treatment association to cause bias. At $\rho_Y = 0$ the threshold is near the feasibility boundary for all DGPs, because a confounder that is irrelevant for the outcome cannot bias the treatment effect estimate regardless of its association with treatment.

The comparison between DGPs 1 and 4 is the most relevant for practitioners. Their thresholds are similar at every ρ_Y . However, while DGP 1 has a true positive treatment effect DGP 4 has a true AME of zero. The threshold table cannot distinguish these two cases. A spurious result with a moderate t -statistic looks identical to a true, signif-

icant result. This is the main limitation of significance-based fragility thresholds which motivates the benchmark analysis in Cinelli et al. (2019) and in Section 4.1.

Note further that in DGPs 1, 2, and 3, the true confounder has a positive effect on the outcome ($\delta_Z > 0$) and a positive association with treatment ($\rho_D^{\text{true}} > 0$), but the true treatment effect is also positive. At the row of the table closest to the true ρ_Y of each DGP, the reported explain-away threshold ρ_D^* exceeds the true ρ_D^{true} . This is expected because although the true confounder exists and creates bias, it is not strong enough to fully explain away the result (since there is a non-trivial treatment effect left after adjusting for it). The method correctly identifies that confounding at the true level is insufficient to overturn the conclusion. Conversely, in DGP 4, the true treatment effect is zero and the short-model AME is an artifact. At the row closest to the true ρ_Y , the reported threshold ρ_D^* falls below the true ρ_D^{true} . This means the true confounder is strong enough to explain away the result. Finally, note that across all four DGPs, $|\rho_D^+| > |\rho_D^*|$, as expected, given that a sign reversal always requires strictly more confounding than statistical insignificance regardless of which direction is examined.

5.2 Study 2: Benchmark validation

We next evaluate the benchmark procedure, using the same DGPs as above. We begin by verifying the correct functioning of the procedure by using the true confounder Z as the benchmark variable. The procedure should then correctly identify the empirical confounding strength of Z and the implied adjusted AME should recover the true value. Table 3 reports results for all four DGPs.

In DGP 1, (where Z has no effect on Y) the Z benchmark yields $\hat{\rho}_{Y,Z} = -0.002$ and an implied bias of 0.003. The adjusted AME of 0.107 is essentially equal to the short-model estimate of 0.110, correctly indicating negligible confounding despite the strong treatment association ($\hat{\rho}_{D,Z} = 0.495$). In DGP 2, the Z benchmark gives $\hat{\rho}_{D,Z} = 0.298$ and $\hat{\rho}_{Y,Z} = 0.095$, consistent with the DGP design ($\rho_D^{\text{true}} = 0.3$, small outcome effect $\delta_Z = 0.2$). The implied bias (correction) of 0.034 accounts for roughly one quarter of the short-model AME of 0.134. The adjusted t -statistic of 14.1 confirms the result survives

Table 3: Benchmark analysis by DGP ($n = 20,000$).

DGP	Var.	$\hat{\rho}_D$	$\hat{\rho}_Y$	ME	SE	t	Correction
1	X_1	0.233	0.144	0.074	0.007	10.57	0.036
	X_2	-0.005	0.081	0.111	0.007	16.35	-0.001
	Z	0.495	-0.002	0.107	0.008	13.51	0.003
2	X_1	0.224	0.136	0.100	0.007	14.52	0.034
	X_2	-0.004	0.101	0.134	0.007	20.13	-0.001
	Z	0.298	0.095	0.100	0.007	14.10	0.034
3	X_1	0.237	0.077	0.243	0.006	40.58	0.022
	X_2	0.015	0.082	0.264	0.006	46.37	0.001
	Z	0.603	0.222	0.057	0.008	7.43	0.208
4	X_1	0.232	0.089	0.102	0.007	14.57	0.021
	X_2	-0.015	0.087	0.124	0.007	18.44	-0.002
	Z	0.401	0.293	-0.014	0.007	-1.98	0.137

ME is the adjusted AME evaluated at $(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k})$. "correction" is defined as $\widehat{\text{AME}}_{\text{short}} - \widehat{\text{AME}}(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k})$. The true AMEs are 0.117 (DGP 1), 0.115 (DGP 2), 0.06 (DGP 3) and 0 (DGP 4). Short-model AMEs are: 0.110 (DGP 1), 0.134 (DGP 2), 0.265 (DGP 3), 0.123 (DGP 4). Z denotes the true omitted confounder included only for validation.

confounding at Z 's empirical strength. Comparing $\hat{\rho}_{D,Z} = 0.298$ to the explain-away threshold at $\rho_Y \approx 0.1$ from Table 2 ($\rho_D^* = 0.641$), the benchmark point lies well inside the robust region.

DGP 3 and DGP 4 provide the strongest validation of the method. In DGP 3, the Z benchmark gives $\hat{\rho}_{D,Z} = 0.603$ and $\hat{\rho}_{Y,Z} = 0.222$, both closely matching the DGP design values ($\rho_D^{\text{true}} = 0.6$, $\delta_Z = 0.8$). The correction is 0.208, reducing the adjusted AME from 0.265 to 0.057, which is close to the true AME of 0.061 from Table 1. The adjusted t -statistic of 7.43 confirms the result is still statistically significant at the true confounding strength. This is consistent with the explain-away threshold at $\rho_Y = 0.2$ from Table 2: $\rho_D^* = 0.678$ for DGP 3, and since $\hat{\rho}_{D,Z} = 0.603 < 0.678$, the benchmark correctly predicts the result is not explained away even by the true confounder.

In DGP 4, the Z benchmark gives $\hat{\rho}_{D,Z} = 0.401$ and $\hat{\rho}_{Y,Z} = 0.293$, closely matching the DGP design ($\rho_D^{\text{true}} \approx 0.4$, $\delta_Z = 0.8$). The correction is 0.137, reducing the adjusted AME from 0.123 to -0.014 with $t = -1.98$ (essentially losing statistical significance at 5% -up to some statistical noise). The adjusted AME thus indicates that the significant short-model result is spurious. Comparing $\hat{\rho}_{D,Z} = 0.401$ to the explain-away threshold at

$\rho_Y \approx 0.3$ from Table 2 ($\rho_D^* = 0.331$), the benchmark coordinates exceed the threshold, correctly predicting that a confounder as strong as Z would explain away the result which the adjusted $t = -1.98$ directly confirms.

In practice, the researcher does not observe the true confounder and must rely on X_1 and X_2 as benchmarks. In our DGPs, X_2 is orthogonal to treatment by construction and is therefore a poor benchmark. Unsurprisingly, then, after partialling out X_1 , $\hat{\rho}_{D,X_2}$ is close to zero in all four DGPs ($-0.005, -0.004, 0.015, -0.015$ respectively). The covariate X_2 does not yield any corrections, regardless of $\hat{\rho}_{Y,X_2}$. Even though X_2 has a moderate association with the outcome residual ($\hat{\rho}_{Y,X_2} \approx 0.08$ – 0.10 across DGPs), omitting it would not cause confounding bias because it is not related to treatment.

The covariate X_1 is moderately correlated with treatment in all four DGPs ($\hat{\rho}_{D,X_1} \approx 0.22$ – 0.24) and also with the outcome residual ($\hat{\rho}_{Y,X_1} \approx 0.08$ – 0.14). In low confounding scenarios (DGPs 1 and 2) X_1 benchmark correctly indicates that confounding at its level would not overturn the result. The adjusted t -statistics remain significant (10.6 and 14.5 respectively). In DGP 2, the X_1 benchmark gives almost identical results to the Z benchmark (bias 0.034 in both cases). In DGP 3, the X_1 benchmark leaves the result highly significant ($t = 40.6$, bias $= 0.022$), while the true confounder Z has $\hat{\rho}_{D,Z} = 0.603$ and $\hat{\rho}_{Y,Z} = 0.222$, far stronger than X_1 on both dimensions. Note, however, that X_1 's benchmark is weak so it could only generate a small correction regardless of the true confounding. The conclusion that the result survives is therefore correct, but weakly motivated/justified.

Once again, DGP 4 is the most important case, as it illustrates the limitations of observed-covariate benchmarking. The X_1 and X_2 benchmarks yield adjusted t -statistics of 14.6 and 18.4 respectively thus suggesting the result is robust to confounding at the level of any observed control. A researcher with access only to X_1 and X_2 as benchmarks would have no reason to doubt the result. In other words, if the true confounder has a stronger association with the outcome than any observed control, then the benchmark may fail to detect confounding.

The results obtained in DGP 3 and DGP 4 simply emphasise the frontiers of sensitivity

analysis in general. A benchmark can correctly signal robustness while still understating the extent of confounding-driven inflation (DGP 3), or fail to flag a spurious result entirely if no observed covariate approaches the true confounder’s strength (DGP 4). From this we conclude, first, that subject-matter knowledge is critical to assess whether plausible omitted variables could be stronger confounders than any covariate in the model and, second, that benchmarking provides reliable bias correction only when the true confounder is no stronger than the best available benchmark on both the treatment and outcome margins simultaneously. When the true confounder exceeds the observed benchmarks, the benchmark-implied correction will understate the bias, and the result may appear more robust than it truly is.

5.3 Study 3: Comparison with Cinelli–Hazlett

Section 2 showed that the nonlinear OVB may be approximated by the Cinelli and Hazlett (2020) framework in low confounding settings, and we discussed an adapted version of that approach inspired by the slope-coefficient interpretation of the leading term of the OVB. We next evaluate the relative performance of the synthetic confounder method compared to the direct application of the Cinelli–Hazlett framework (CH-OLS) and to the adapted version (CH-adapted) described in Section 2.

Cinelli and Hazlett (2020) show that, in a linear model, the bias in the short-model OLS estimator due to omitting Z can be written as a function of $R_{D \sim Z | \mathbf{X}}^2$ (the share of residual treatment variation explained by Z), and $R_{Y \sim Z | D, \mathbf{X}}^2$ (the share of residual outcome variation explained by Z after conditioning on treatment). The bias formula is

$$|\text{bias}| = \sqrt{\frac{R_{Y \sim Z | D, \mathbf{X}}^2 \cdot R_{D \sim Z | \mathbf{X}}^2}{1 - R_{D \sim Z | \mathbf{X}}^2}} \cdot \widehat{\text{SE}}(\hat{\beta}^*) \cdot \sqrt{df}, \quad (5.2)$$

and the adjusted estimate is $\hat{\beta}_{\text{adj}}^* = \hat{\beta}^* - \text{sign}(\hat{\beta}^*) \cdot |\text{bias}|$. In the linear model, $R_{D \sim Z | \mathbf{X}}^2 = \rho_{D,Z}^2$ and $R_{Y \sim Z | D, \mathbf{X}}^2 = \rho_{Y,Z}^2$, where $\rho_{D,Z}$ and $\rho_{Y,Z}$ denote the partial correlations of Z with the treatment and outcome residuals after partialling out $\mathbf{X}$. For nonlinear models, Lemma 2.1 shows that an additional remainder $R = O(\delta_0^2)$ enters the bias expression,

so (5.2) becomes a first-order approximation whose error grows quadratically with confounding strength.

Because Z is not observed in practice, a researcher specifies hypothetical values of the two R^2 quantities and benchmarks them against the partial R^2 of observed covariates to assess whether such values are plausible. In the present simulation, however, Z is observed by construction, which allows us to compute the true benchmark correlations $\hat{\rho}_{D,Z}$ and $\hat{\rho}_{Y,Z}$ in each replication. We can then let $R_{DZ}^2 = \hat{\rho}_{D,Z}^2$ and $R_{YZ}^2 = \hat{\rho}_{Y,Z}^2$ and feed these quantities to all three methods. It would follow that any difference in the quality of the adjusted AME across methods is attributable to the methods themselves, rather than benchmarking.

Specifically, the methods are as follows:

1. In the standard Cinelli-Hazlett method (CH-OLS) the binary outcome is approximated by a linear probability model of Y on D and $\mathbf{X}$, computes OLS residuals for Y and D , sets $R_{DZ}^2 = \hat{\rho}_{D,Z}^2$ and $R_{YZ}^2 = \hat{\rho}_{Y,Z}^2$ from those residuals, and plugs into (5.2) to obtain an adjusted coefficient.
2. The adapted Cinelli-Hazlett (CH-adapted) fits the short logit model, computes $\widehat{\text{AME}}_{\text{short}}$ and $\widehat{\text{se}}_{\text{short}}$, and replaces OLS residuals with logit standardised residuals $\tilde{Y}_i = (Y_i - \hat{\mu}_i) / \sqrt{\hat{\mu}_i(1 - \hat{\mu}_i)}$ when computing R_{DZ}^2 and R_{YZ}^2 . These partial R^2 values and the logit AME are then plugged into (5.2),

$$\widehat{\text{AME}}_{\text{adj}}^{\text{CH-adapted}} = \widehat{\text{AME}}_{\text{short}} - \text{sign}(\widehat{\text{AME}}_{\text{short}}) \sqrt{\frac{\hat{\rho}_{Y,k}^2 \cdot \hat{\rho}_{D,k}^2}{1 - \hat{\rho}_{D,k}^2}} \cdot \widehat{\text{se}}_{\text{short}} \cdot \sqrt{df}. \quad (5.3)$$

3. The synthetic confounder uses the same standardised residuals and the same $(\hat{\rho}_{D,Z}, \hat{\rho}_{Y,Z})$ as CH-adapted, but replaces the linear bias formula with the procedure described in Section 3.

For this experiment, we use DGP (5.1) with parameter values $\alpha = -0.5$, $\gamma_1 = 0.3$, $\gamma_2 = 0.2$. The key regressor takes values in $\beta_D \in \{0, 0.5\}$. For each value of β_D we consider three levels of confounding strength, $\delta_Z \in \{0.2, 0.5, 0.8\}$ with corresponding

$\rho_D^{\text{true}} \in \{0.3, 0.4, 0.6\}$ respectively, so that stronger outcome confounding is paired with stronger treatment selection. We generated 10,000 samples of 5,000 observations. For each sample we computed the adjusted AMEs and we report the mean of the absolute errors, $|\widehat{\text{AME}}_{\text{adj}} - \text{AME}_{\text{true}}|$ (where AME_{true} is computed from the correctly specified logit on a simulated sample of $n = 500,000$ draws).

Table 4: Adjusted AME accuracy by confounding strength ($n = 5,000, 10,000$ simulations)

Panel	(δ_Z, ρ_D)	MAE			Ratio / Synthetic	
		CH-OLS	CH-adapted	Synthetic	CH-OLS	CH-adapted
A: $\beta_D = 0$	(0.2,0.3)	0.0115	0.0114	0.0121	0.94	0.94
	(0.5,0.4)	0.0141	0.0145	0.0129	1.09	1.12
	(0.8,0.6)	0.0479	0.0547	0.0151	3.17	3.63
B: $\beta_D = 0.5$	(0.2,0.3)	0.0121	0.0113	0.0117	1.03	0.96
	(0.5,0.4)	0.0177	0.0170	0.0123	1.45	1.38
	(0.8,0.6)	0.0651	0.0723	0.0143	4.54	5.04

Notes: MAE = mean|adjusted AME – true AME|. CH-OLS applies the Cinelli and Hazlett (2020) formula to the short OLS (LPM) model and OLS residuals. CH-adapted applies the same formula to logit standardised residuals and the logit AME. Synthetic re-estimates the full logit at the benchmark $(\hat{\rho}_{D,Z}, \hat{\rho}_{Y,Z})$. Ratio = CH MAE / Synthetic MAE; values above 1 favour the synthetic method.

The results are presented in Tables 4 and 5. When confounding is weak ($\delta_Z = 0.2$) the nonlinear remainder is small and the CH-OLS provides a good approximation. Table 4 reports the ratio of Cinelli-Hazlett MAE to Synthetic MAE, which equals 0.94 when $\beta = 0$ and 1.03 when $\beta = 0.50$. As we introduce confounding (δ_Z increases) the ratio also increases. For instance, when $\beta = 0.5$ it rises from 1.03 at $\delta_Z = 0.2$ to 1.45 at $\delta_Z = 0.5$ and 4.54 at $\delta_Z = 0.8$. Similar increases are seen when $\beta_D = 0$. Table 5 shows that when the Cinelli–Hazlett bias grows roughly sixteen-fold as we increase confounding four-fold from $\delta_Z = 0.2$ to $\delta_Z = 0.8$. This is consistent with the sixteen-fold increase implied by the nonlinear OVB formula.

Table 5 reports the signed bias underlying the MAE figures in Table 4. Both Cinelli–Hazlett variants under-correct systematically and increasingly as δ_Z grows, in both the null panel ($\beta_D = 0$) and the positive-effect panel ($\beta_D = 0.5$). This confirms that the MAE differences in Table 4 are driven by bias rather than increased variance, and is consistent with the $O(\delta_0^2)$ nonlinear remainder term identified in Section 2. That is, the CH correction, being a first-order approximation, fails to absorb the growing second-order

Table 5: Signed bias by confounding strength ($n = 5,000, 10,000$ simulations)

Panel	(δ_Z, ρ_D)	CH-OLS	CH-adapted	Synthetic
A: $\beta_D = 0$	(0.2,0.3)	0.0027	0.0027	0.0003
	(0.5,0.4)	0.0091	0.0099	0.0008
	(0.8,0.6)	0.0479	0.0547	0.0012
B: $\beta_D = 0.5$	(0.2,0.3)	0.0038	0.0015	-0.0011
	(0.5,0.4)	0.0153	0.0146	-0.0000
	(0.8,0.6)	0.0651	0.0723	0.0021

Notes: Bias = mean(adjusted AME – true AME); positive values indicate overcorrection, negative values indicate undercorrection. CH-OLS and CH-adapted bias grows with δ_Z in both panels, consistent with the $O(\delta_0^2)$ nonlinear remainder term. The synthetic method’s bias remains close to zero throughout, including in Panel A where the true treatment effect is zero.

component of the bias. Finally, the standard and adapted Cinelli–Hazlett approaches perform similarly throughout, so that replacing OLS residuals with logit standardised residuals does not improve accuracy.

6 Empirical Illustration: Cognitive Skills and Labour Supply in the 1970 British Cohort Study

In recent decades, a large body of research has highlighted the importance of cognitive skills for economic and social outcomes, including educational attainment, employment, earnings, occupational status, health, and social mobility (Heckman et al., 2006; Hanushek and Woessmann, 2008¹⁰). This research has motivated substantial interest in understanding how cognitive skills develop over the life course and the extent to which cognitive abilities acquired during childhood have lasting effects on outcomes in adulthood. The latter matters because a substantial share of cognitive ability appears to be established early in life, with many cognitive skills exhibiting considerable persistence from late childhood onwards, even after discounting the effect of noncognitive skills (Heckman et al., 2013; Cunha and Heckman, 2007)

¹⁰Several mechanisms may explain these long-run associations. Higher cognitive skills facilitate learning, the acquisition of greater educational attainment and advanced skills. Cognitive skills may also directly enhance productivity by improving individuals’ ability to process information, solve problems, and adapt to changing environments. Together, these mechanisms may generate positive returns in the labour market through higher employment probabilities, greater earnings, and access to occupations of higher quality.

Estimating the causal effect of cognitive skills on labour market outcomes is, however, challenging because cognitive ability is endogenous to many of the outcomes researchers seek to explain (such as employment experiences, occupational environments, and educational investments). To address these concerns, researchers have frequently relied on longitudinal cohort studies such as the British Cohort Study (BCS70), which measure skills during childhood or adolescence, well before adulthood outcomes of interest are realised (Del Bono et al., 2024; Golsteyn et al., 2014; Castillo et al., 2019; Angerer et al., 2023; Denny and Doyle, 2008; Castillo et al., 2011; Deary et al., 2000; Schoon et al., 2010; Gill and Prowse, 2024; Fé and Jahangir, 2026). Because cognitive assessments are collected prior to most adult educational, occupational, and family choices, they provide a plausibly exogenous measure of skills and reduce concerns regarding biases due to reverse causality and omitted variables. However, they do not fully eliminate the risk, making sensitivity analyses an important step to consolidate evidence.

This section of the paper studies the robustness of the association between childhood cognitive skills and labour market outcomes decades later, once a large set of early-life observable confounders is taken into account. The study uses the BCS70, which follows everyone born in Great Britain in a single week of April 1970 (approximately 17,000 births) with follow-up sweeps which, at the time of writing, cover ages 5, 10, 16, 26, 30, 34, 38, 42, 46, and 51. For this illustration, the paper focuses on labour market outcomes at the age 46 wave. Specifically, the analysis uses binary indicators of employment and managerial jobs, which are proxies of labour force participation and the quality of jobs. The specification of the short model is,

$$\mathbb{E}(Y_i \mid D_i, \mathbf{X}_i) = \Lambda(\alpha + \beta D_i + \mathbf{X}_i' \boldsymbol{\gamma}), \quad (6.1)$$

where $\Lambda(\cdot)$ is the logistic CDF. In the absence of a sensitivity analysis, a researcher would be implicitly assuming Conditional Independence, by which

$$\mathbb{E}(Y_i \mid D_i, \mathbf{X}_i, Z_i) = \mathbb{E}(Y_i \mid D_i, \mathbf{X}_i) \quad \text{for all } Z_i, \quad (6.2)$$

that is, conditional on the observed covariates $\mathbf{X}_i$, no unobserved variable Z_i exists that is simultaneously associated with cognitive skills D_i and with the outcome Y_i . Equation (6.2) is equivalent to assuming that any such Z_i satisfies $\rho_D = \text{Cor}(Z_i, \tilde{D}_i) = 0$ or $\rho_Y = \text{Cor}(Z_i, \tilde{Y}_i) = 0$, where $\tilde{D}_i$ and $\tilde{Y}_i$ are the residualised treatment and outcome defined in Section 3.

The key treatment variable, D , is an index of cognitive skills assessed at age 10¹¹. Cognitive ability at age 10 was measured via the Friendly Maths Test, comprising 51 applied numerical-reasoning problems, together with four subtests of the British Ability Scales¹². Because test administration spanned several months in 1980, each component score was first age-standardised within two-month (bimester) bins defined by the child’s exact age at assessment. As in (Gill and Prowse, 2024; Fé and Jahangir, 2026), the five age-standardised scores were then combined via principal component analysis, retaining the first principal component as a single summary measure of cognitive skill.

The vector $\mathbf{X}$ includes a rich collection of regressors. Perinatal controls include the mother’s age at delivery (and its square), region of birth, maternal smoking during pregnancy, gestational abnormality, infant sex, birth weight, parity, and prior pregnancy losses. Age-5 controls include breastfeeding, inpatient hospital episodes, sensory or speech difficulties, whether the child was read to, parental qualifications and employment status, ethnicity, and number of siblings. Age-10 controls include further inpatient episodes, parental unemployment and benefit receipt, household income band, housing tenure, number of rooms, parental smoking, sensory or speech difficulties, standardised height and weight, and teacher-reported disengagement.

The vector $\mathbf{X}$ further includes teacher-reported measures of the child’s levels of developmental behaviour at age 10. This assessment is based on a multiple item questionnaire that captures various aspects of the child’s character. From the questionnaire we keep a raw measure of perseverance (Fé and Jahangir (2026)), alongside six principal com-

¹¹The full description of the measure and its construction is given in Fé and Jahangir (2026)

¹²The BAS included the following tasks: Word Definitions (vocabulary and verbal comprehension), Word Similarities (abstract verbal reasoning), Digit Recall (short-term and working memory), and Progressive Matrices (nonverbal pattern recognition, analogous to Raven’s Progressive Matrices)

ponents summarising developmental behaviours orthogonal to perseverance¹³. The first component “Emotional Sensitivity and Social Dysfunction”, contrasts aggressive social behaviours such as teasing and bullying against emotional vulnerability indicators like nervousness and clumsiness. The second component, “Anxiety and Fine Motor Difficulties”, combines anxiety-related behaviours with fine motor skill challenges, potentially reflecting developmental coordination difficulties. The remaining components capture more specific dimensions: Component 3 relates to incontinence and physical coordination problems; Component 4 reflects a contrast between impulsivity and social withdrawal; Component 5 combines motor skill deficits with stereotypic behaviours such as tics and repetitive vocalisations, possibly associated with neurodevelopmental conditions; and Component 6 emphasises severe motor coordination difficulties.

Table 6 reports average marginal effects (AMEs) for both outcomes, restricted to covariates significant at the 10% level or better. Cognitive skills predict both outcomes strongly and significantly. A one-unit increase in the index of cognitive skills raises the probability of full-time employment by 2.7 percentage points and the probability of holding a managerial or supervisory position by 2.9 percentage points.

6.1 Sensitivity Analysis

We applied the synthetic confounder method to the short model. The fragility thresholds¹⁴ are presented in Table 7. At $\rho_Y = 0$ the thresholds sit near the boundary of the feasibility region in Proposition 3.1 (0.871 for full-time employment, 0.621 for managerial status) as we expected, since a confounder that does not move the outcome at all cannot bias the estimate. As we move away from zero, however, the threshold falls quickly. By $\rho_Y = \pm 0.1$ (a confounder explaining a 1% of the outcome residual’s variance) the explain-away threshold has already dropped to 0.333 for full-time employment and 0.217 for managerial status.

The table also shows that fragility depends on the direction of the hypothetical con-

¹³The construction of these components is discussed in Fé and Jahangir (2026).

¹⁴Recall from Section 3 that we may read the results as follows. A $\rho_Y = 0.1$ is a correlation, and the corresponding variance share is ρ_Y^2 , so $\rho_Y = 0.1$ corresponds to a confounder explaining only 1% of the variance in the outcome residual by outcome

Table 6: Average marginal effects on full-time employment and managerial status at age 46 (significant effects only).

	Full-time employment		Manager/Supervisor	
	AME	SE	AME	SE
Cognitive skills	0.0267***	(0.0062)	0.0291***	(0.0083)
Female	−0.4061***	(0.0151)	−0.1466***	(0.0225)
Father unemployed at age 10	0.1186***	(0.0375)	0.1515**	(0.0738)
Behavioural PC6	0.0175**	(0.0086)	—	—
Behavioural PC1	—	—	0.0199***	(0.0058)
Region 5 ^a	−0.0901***	(0.0344)	—	—
Region 6	−0.1253**	(0.0504)	—	—
Region 7	—	—	0.0903*	(0.0468)
Mother has no qualifications	0.0538*	(0.0318)	—	—
Mother stays home at age 10	−0.0337*	(0.0203)	—	—
Mother stayed in education beyond SLA	—	—	0.0707**	(0.0283)
Child was read to at age 5	—	—	0.0599**	(0.0255)
Non-white ethnicity	—	—	0.1058**	(0.0429)
Standardised height at age 10	—	—	0.0464***	(0.0150)
Standardised weight at age 10	—	—	−0.0323**	(0.0149)
Mother works full-time at age 5	—	—	−0.0589*	(0.0313)
Income band 1 (lowest)	—	—	−0.1064*	(0.0600)
Housing tenure category 2	—	—	−0.3822*	(0.1990)
Housing tenure category 3	—	—	−0.3536*	(0.1940)
Housing tenure category 4	—	—	−0.3370**	(0.1532)
Housing tenure category 5	−0.5166**	(0.2621)	—	—
Observations	3,584		2,420	
Pseudo R^2	0.181		0.065	

^a Region indicators are coded relative to the omitted reference category (Region 10).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Fragility thresholds by outcome and ρ_Y , BCS70 age-46 labour supply.

ρ_Y	Full-time employment		Manager/Supervisor	
	ρ_D^*	ρ_D^+	ρ_D^*	ρ_D^+
-0.5	-0.104	-0.198	-0.126	-0.248
-0.4	-0.115	-0.223	-0.135	-0.278
-0.3	-0.139	-0.279	-0.157	-0.349
-0.2	-0.201	-0.442	-0.215	-0.603
-0.1	-0.449	robust	-0.484	robust
0.0	0.871	robust	0.621	0.931
0.1	0.333	0.515	0.217	0.449
0.2	0.202	0.328	0.123	0.277
0.3	0.150	0.241	0.086	0.197
0.4	0.125	0.193	0.070	0.153
0.5	0.112	0.166	0.061	0.126

ρ_D^* is the explain-away threshold; ρ_D^+ is the sign-change threshold. “robust” indicates that no threshold is reached within the feasible range. Short-model AMEs: 0.0233 (SE 0.0054) for full-time employment; 0.0266 (SE 0.0075) for managerial/supervisory status.

founder. For $\rho_Y < 0$ (a confounder that would suppress the outcome net of cognition) full-time employment’s threshold is smaller in magnitude than managerial status’ at every row (e.g. -0.104 versus -0.126 at $\rho_Y = -0.5$), making employment the more fragile outcome in this direction. For $\rho_Y > 0$, managerial status’ threshold is smaller at every row (0.217 versus 0.333 at $\rho_Y = 0.1$; 0.123 versus 0.202 at $\rho_Y = 0.2$; and so on), making managerial status the more fragile outcome. This second direction is also the more plausible one to worry about in this setting because the candidate omitted confounders will include family socioeconomic advantage, parental ability, or early-childhood investment, which are factor expected to raise both childhood cognitive skill and adult labour-market success simultaneously¹⁵.

¹⁵To see why, consider the situation when $\rho_D > 0, \rho_Y > 0$ where the confounder Z refers to, for example, a family advantage not captured by the controls. $\rho_D > 0$ implies that more advantaged children tend to score higher on the cognitive measure (net of controls) and $\rho_Y > 0$ implies that more advantaged adults tend to do somewhat better in the labour market for reasons unrelated to cognition. In the short model, which omits Z , individuals with high cognitive scores also tend to have high Z (since $\rho_D > 0$), and those same individuals would have done well in the labour market regardless of their cognition, simply because they are also advantaged on Z (since $\rho_Y > 0$). Part of what the short model attributes to cognition is therefore due to Z . Adding Z back into the model absorbs this portion of the association, shrinking the adjusted AME toward zero. By contrast, when ρ_D and ρ_Y have opposite signs, the confounder pulls the bias in the opposite direction. For instance if $\rho_D < 0, \rho_Y > 0$, individuals with high cognition tend to have low Z , but low Z here predicts a worse outcome, so a short model’s positive AME would be understated, and adding Z back would push the adjusted AME further from zero rather

In general, the explain-away threshold ρ_D^* never reaches “robust” status at any ρ_Y in the range considered for either outcome. Thus, we conclude that a sufficiently strong confounder always exists that would drive cognitive skills’ association with labour-market status to statistical insignificance. For full-time employment, sign reversal is infeasible within the scanned grid at both $\rho_Y = -0.1$ and $\rho_Y = 0.0$, while for managerial status, only $\rho_Y = -0.1$ shows this.

As explained in Section 4.1, it is possible to tie the choice of ρ_Y to the current application at hand by asking how much would the estimate shift if a confounder as strong as a specific observed variable had been omitted. In principle, the choice of benchmarks can be informed by the regression results in Table 6. The largest marginal effect in either model belongs to sex, with women being significantly less likely to be in full-time employment or to hold a managerial position than men. Father’s unemployment at age 10 is also a strong significant predictor of both outcomes. Therefore, on the basis of their strong predictive power, the benchmarking analysis could rely on sex, father’s unemployment status at age 10 and whether the mother stayed at home at age 10 as benchmarks. However, our main concern is the existence of confounder which are not orthogonal to treatment (those which are do not bias the estimates of the treatment effect).

Table 8, which evaluates the fragility of the results against the selected benchmarks, reveals that none of the previous predictors are strongly associated with cognitive skills. Specifically, for sex, father’s unemployment and mother staying home at age 10, $|\hat{\rho}_D| \leq 0.095$. For the latter two benchmarks, this falls below the corresponding explain-away thresholds, but because of the weak association with the treatment, the implied bias should be small (and this is confirmed by the results).

Given this, the paper considers a second set of benchmarks based on the largest partial correlations with cognitive skills, presented in Table 9 in Appendix D. In particular, the selected benchmarks are age standardised weight and height and the first behavioural principal component (PC1), capturing emotional sensitivity and social dysfunction. Table 8 shows how these predictors have, in general, strong associations with the residual

than toward it.

Table 8: Benchmark analysis: strongest observed predictors, BCS70 age-46 labour supply.

FT Employment

Variable	$\hat{\rho}_D$	$\hat{R}_D^2$	$\hat{\rho}_Y$	$\hat{R}_Y^2$	ME	t	Bias
Sex	-0.074	0.55%	-0.399	15.92%	0.0138	2.76	0.0095
Mother stay home	-0.009	0.01%	-0.032	0.10%	0.0230	4.30	0.0003
Father unemployed	-0.012	0.01%	0.049	0.24%	0.0239	4.46	-0.0006
PC1	0.168	2.82%	0.004	0.00%	0.0228	4.21	0.0005
Height at 10	0.134	1.80%	0.026	0.07%	0.0221	4.09	0.0012
Weight at 10	0.082	0.67%	-0.030	0.09%	0.0238	4.44	-0.0005

Manager

Variable	$\hat{\rho}_D$	$\hat{R}_D^2$	$\hat{\rho}_Y$	$\hat{R}_Y^2$	ME	t	Bias
Sex	-0.095	0.90%	-0.130	1.69%	0.0236	3.14	0.0030
Mother stay home	-0.006	0.00%	-0.023	0.05%	0.0266	3.52	-0.0000
Father unemployed	-0.008	0.01%	0.038	0.14%	0.0263	3.49	0.0003
PC1	0.161	2.59%	0.069	0.48%	0.0202	2.63	0.0064
Height at 10	0.115	1.32%	0.063	0.40%	0.0221	2.91	0.0044
Weight at 10	0.105	1.10%	-0.045	0.20%	0.0276	3.63	-0.0010

ME is the adjusted AME evaluated at $(\hat{\rho}_D, \hat{\rho}_Y)$. Bias = ME_{short} - ME_{adjusted}. $\hat{R}_D^2 = \hat{\rho}_D^2$ and $\hat{R}_Y^2 = \hat{\rho}_Y^2$ are the implied shares of residual (post-control) treatment and outcome variance respectively.

variation in treatment (in the 1% - 2.8% range). Note that although these values are modest in absolute terms, this range is typical for partial R^2 after a rich control set.

Of every benchmark considered, PC1 comes closest to its explain-away threshold ρ_D^* , that is, its treatment correlation $\hat{\rho}_D$ gets close to the point at which a confounder of that strength would render the result statistically insignificant. For full-time employment, PC1 has coordinates $(\hat{\rho}_D, \hat{\rho}_Y) = (0.168, 0.004)$. The nearest row of Table 7 ($\rho_Y = 0.0$) has an explain away threshold of $\rho_D^* = 0.871$, leaving a wide margin of about 80% ($0.168/0.871 \approx 0.19$). For managerial status, however, the coordinates are $(0.161, 0.069)$, and the nearest threshold ($\rho_D^* = 0.217$ at $\rho_Y = 0.1$) leaves a margin of about 25%.

Overall, confounding as strong as the strongest benchmarks considered do not explain away the association between cognitive skills and managerial status. However, we understand that a confounder only modestly stronger than the PC1 would be sufficient to overturn the conclusion. The association with full time employment remains robust to confounding at the strength of any of the benchmarks considered.

7 Conclusion

This paper has developed a sensitivity analysis framework for likelihood-based nonlinear models, extending Cinelli and Hazlett (2020)’s approach for linear regression. The paper first discusses how the omitted variable bias of a likelihood-based estimator contains a non-vanishing remainder term that grows quadratically with the strength of the confounder and depends on the curvature of the model’s nonlinearity. The implication is that sensitivity methods designed for linear models can be applied directly to nonlinear models, but only as a first-order approximation that performs well under weak confounding. Under moderate to strong confounding, such methods struggle to recover the true magnitude of the bias.

In view of this result, the paper proposes a simulation-based synthetic confounder approach. Rather than deriving a closed-form bound and the corresponding correction, the method constructs a synthetic confounder with exact, pre-specified partial correlations

with the residualised treatment and outcome, and recovers the exact adjusted average marginal effect by re-estimating the long model. Using this method, the paper develops a fragility framework that allows researchers to compute, for any given outcome-residual correlation, the explain-away and sign-change thresholds. That is, the fragility framework charts the strength of confounding required to eliminate the statistical significance of the estimated effect or to reverse its sign.

Because the partial correlations of unobserved confounders are not known in practice, the paper also develops a benchmarking procedure, following Cinelli and Hazlett (2020), that grounds the sensitivity analysis in the specific empirical application. The procedure asks how the result would change if an omitted confounder were as strongly correlated with treatment and outcome as a specific observed covariate. Selecting informative benchmarks still requires subject-matter knowledge, particularly regarding which observed covariates are likely to proxy for the strength of plausible omitted confounders on both margins. The benchmark estimates can then be compared against the fragility thresholds to assess how plausible it is that confounding of the relevant magnitude could overturn the conclusion.

A Monte Carlo simulation evaluates the performance of the proposed procedures and confirms the theoretical predictions of Section 2. In particular, the simulations show that direct application of linear-model sensitivity methods to nonlinear models is only a first-order approximation whose accuracy deteriorates as confounding strengthens, while the synthetic confounder method remains accurate throughout, including under strong confounding.

The method is then applied to assess the robustness of cognitive skills measured at age 10 as a predictor of labour-market outcomes in adulthood. Using data from the 1970 British Cohort Study, the paper reports estimates of short models of labour force participation (full-time employment) and job quality (managerial or supervisory status) that include a composite measure of cognitive skills alongside a rich set of pre-treatment covariates, including parental background and parent- and teacher-reported measures of developmental behaviour. The concern motivating the analysis is that, despite the rich-

ness of this control set, residual sources of confounding arising from parental background, environmental factors, and similar unobserved influences may remain. Such confounders are likely to be positively associated with cognitive skills and adult labour-market outcomes, which would inflate the estimated effect of cognitive skill. Applying the full suite of methods developed in this paper, we find that cognitive skills remain a robust predictor of employment participation, but are a comparatively fragile predictor of job quality: a benchmark only modestly stronger than the strongest observed covariate would be sufficient to overturn the latter conclusion.

The exposition has assumed, for convenience, that the confounder enters the outcome equation homogeneously, without interacting with the treatment; the method itself does not require this, since the synthetic confounder is built directly from the partial correlations (ρ_D, ρ_Y) rather than from a structural coefficient, and treatment-effect heterogeneity can be accommodated by re-estimating the long model with an added interaction term, as discussed in Section 3.

The framework presented here has a number of limitations that point to directions for future research. In a recent paper, Diegert et al. (2025) develop axiomatic results for sensitivity frameworks. Extending that axiomatic treatment to the simulation-based setting is a natural but non-trivial next step, and is left for future research. As in Cinelli and Hazlett (2020), the partial correlations are fixed to levels decided by the researcher, and one might wish to allow for uncertainty in these choices, allowing for sampling variability around the sensitivity parameters themselves. The framework has also been developed with a single confounder in mind. In practice, multiple confounders may operate simultaneously. The single synthetic confounder can, however, be interpreted as a summary of their combined influence. Furthermore, how to handle multiple unobservable quantities whose interactions with treatment and outcome are complex and potentially distinct from one another remains an open question. Despite these limitations, the synthetic confounder framework offers applied researchers a tractable and flexible tool for assessing the robustness of causal conclusions in the nonlinear models that are routine in empirical economics.

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A Ancillary Results.

The following results are included for completeness and support the proof of the nonlinear OVB. To proceed, the following standard regularity conditions are assumed to hold throughout.

- R1.** The parameter space $\Theta \subset \mathbb{R}^k$ is compact, with both $\boldsymbol{\theta}_0$ and $\boldsymbol{\theta}^*$ being interior points of Θ .
- R2.** The log-likelihoods $\ell_i(\boldsymbol{\theta})$ and $\ell_i^*(\boldsymbol{\theta})$ are twice continuously differentiable in $\boldsymbol{\theta}$ on Θ , almost surely.
- R3.** The regressors W_i satisfy a moment condition sufficient for dominated convergence of the score and Hessian under the model family¹⁶:
- R4. Dominated convergence of the score.** Under condition **R3**, there exists a square-integrable function $g_1(Y_i, W_i)$ with $\mathbb{E}[g_1^2] < \infty$ such that $\|\mathbf{s}_i(\boldsymbol{\theta})\| \leq g_1(Y_i, W_i)$ for all $\boldsymbol{\theta} \in \Theta$.
- R5. Dominated convergence of the Hessian.** Under condition **R3**, there exists an integrable function $g_2(Y_i, W_i)$ with $\mathbb{E}[g_2] < \infty$ such that

$$\|\mathbf{H}_i(\boldsymbol{\theta})\| \leq g_2(Y_i, W_i)$$

for all $\boldsymbol{\theta} \in \Theta$.

- R6.** The population information matrices $\mathcal{I}^*(\boldsymbol{\theta}_0)$ and $\mathcal{I}(\boldsymbol{\theta}^*)$ are both finite and positive definite.

These assumptions are standard in the literature (e.g. White, 1982). In particular, R4 justifies differentiating under the expectation when deriving the properties of the score,

¹⁶The strength of the assumptions imposed on the covariates varies from model to model. In bounded variance models (such as logit or cloglog) $\mathbb{E}[\|W_i\|^2] < \infty$ is sufficient. In unbounded variance models such as Poisson we require bounded moment generating functions, in which case $\mathbb{E}[e^{c\|W_i\|}] < \infty$ for some $c > 0$ and $\|\mathbf{s}_i(\boldsymbol{\theta})\| \leq (Y_i + e^{C\|W_i\|})\|W_i\|$, $\|\mathbf{H}_i(\boldsymbol{\theta})\| \leq e^{C\|W_i\|}\|W_i\|^2$ are both bounded.

and guarantees that $\mathbf{s}(\boldsymbol{\theta})$ is continuous. Similarly, R5 justifies exchanging differentiation and expectation to obtain:

$$\frac{\partial \mathbf{s}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} = \mathbb{E}[\mathbf{H}_i(\boldsymbol{\theta})] = -\mathcal{I}(\boldsymbol{\theta}), \quad (\text{A.1})$$

and guarantees that $\mathbf{s}(\boldsymbol{\theta})$ is continuously differentiable on Θ . Under these conditions, consistency of $\hat{\boldsymbol{\theta}}^*$ to $\boldsymbol{\theta}^*$ follows from conditions R1-R6 and a uniform law of large numbers White (1982).

To proceed further, we need the following well known results from the Maximum Likelihood Literature. First, the expected value of the score is zero at the true parameter is zero,

$$\mathbb{E}[\mathbf{s}_i(\boldsymbol{\theta}_0)] = \mathbf{0}. \quad (\text{A.2})$$

Second, the short model is the long model with δ restricted to zero and so, for every observation i and every parameter vector $\boldsymbol{\theta}$,

$$\ell_i^*(\boldsymbol{\theta}) = \ell_i(\boldsymbol{\theta}, 0), \quad (\text{A.3})$$

where $\ell_i(\boldsymbol{\theta}, \delta)$ is the long-model log-likelihood and $\ell_i^*(\boldsymbol{\theta})$ is the short-model log-likelihood. Differentiating both sides with respect to $\boldsymbol{\theta}$ gives the same equality for the scores: $\mathbf{s}_i^*(\boldsymbol{\theta}) = \mathbf{s}_i(\boldsymbol{\theta}, 0)$ pointwise. It follows that applying (A.2) to the long model when the true parameter is $(\boldsymbol{\theta}_0, 0)$, all components of the long-model score have expectation zero:

$$\mathbb{E}_0[\mathbf{s}_i(\boldsymbol{\theta}_0, 0)] = \mathbf{0}, \quad (\text{A.4})$$

including the component with respect to δ :

$$\mathbb{E}_0[s_{i,\delta}(\boldsymbol{\theta}_0, 0)] = 0. \quad (\text{A.5})$$

Further, by definition of $\boldsymbol{\theta}^*$ as the solution to $\mathbb{E}[\mathbf{s}_i^*(\boldsymbol{\theta}^*)] = \mathbf{0}$, and using (A.3):

$$\mathbb{E}[\mathbf{s}_i(\boldsymbol{\theta}^*, 0)] = \mathbb{E}[\mathbf{s}_i^*(\boldsymbol{\theta}^*)] = \mathbf{0}. \quad (\text{A.6})$$

In particular, the long-model score for the included parameters $(\boldsymbol{\theta}_A, \beta)$, evaluated at $(\boldsymbol{\theta}^*, 0)$, is zero.

With these results we are now in a position to establish that, in a neighbourhood of $\delta_0 = 0$, the pseudo-true parameter $\boldsymbol{\theta}^*$ is a continuously differentiable function of the confounding strength δ_0 . To do so we apply the Implicit Function Theorem to the short-model population score equation, expanding around the point $(\boldsymbol{\theta}_0, 0)$ where there is no confounding. At this point the short model is correctly specified, so $\boldsymbol{\theta}^* = \boldsymbol{\theta}_0$, and the preliminary results above guarantee that the conditions of the IFT are satisfied. The theorem then delivers both the existence of $\boldsymbol{\theta}^*(\delta_0)$ as a continuous differentiable function and an explicit expression for its derivative at $\delta_0 = 0$, which characterises the direction and first-order rate at which the pseudo-true parameter departs from the truth as confounding is introduced.

Lemma A.1. *Let*

$$F(\boldsymbol{\theta}^*, \delta_0) := \mathbb{E}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}^*)] = \mathbf{0}, \quad (\text{A.7})$$

*where the expectation is taken under the true DGP with confounding strength δ_0 . Under conditions **R1–R6**, there exists a continuously differentiable function $\delta_0 \mapsto \boldsymbol{\theta}^*(\delta_0)$ satisfying $F(\boldsymbol{\theta}^*(\delta_0), \delta_0) = \mathbf{0}$ in a neighbourhood of $\delta_0 = 0$, with $\boldsymbol{\theta}^*(0) = \boldsymbol{\theta}_0$ and*

$$\boldsymbol{\theta}^*(\delta_0) = \boldsymbol{\theta}_0 + \boldsymbol{\Pi} \cdot \delta_0 + O(\delta_0^2), \quad (\text{A.8})$$

where

$$\boldsymbol{\Pi} := [\mathcal{I}^*(\boldsymbol{\theta}_0)]^{-1} \text{Cov}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)]. \quad (\text{A.9})$$

Here $\mathcal{I}^(\boldsymbol{\theta}_0) := -\mathbb{E}_{\delta_0=0}[\mathbf{H}_i^*(\boldsymbol{\theta}_0)]$ is the short-model information matrix evaluated at the true parameter $\boldsymbol{\theta}_0$ where the short model is correctly specified. Thus, $\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*(\delta_0)\| = O(\delta_0)$ as $\delta_0 \rightarrow 0$.*

Proof. We apply the Implicit Function Theorem (IFT) to $F(\boldsymbol{\theta}^*, \delta_0) = \mathbf{0}$ at the point $(\boldsymbol{\theta}_0, 0)$. Our first step is to establish that, indeed, the latter point belongs to the contour of F , so that F is zero at $(\boldsymbol{\theta}_0, 0)$. At $\delta_0 = 0$ the short and long models coincide by (A.3), so $\mathbf{s}_i^*(\boldsymbol{\theta}_0) = \mathbf{s}_i(\boldsymbol{\theta}_0, 0)$ pointwise. Since the long model is correctly specified at $(\boldsymbol{\theta}_0, 0)$, score orthogonality (A.2) gives:

$$F(\boldsymbol{\theta}_0, 0) = \mathbb{E}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}_0)] = \mathbb{E}_{\delta_0=0}[\mathbf{s}_i(\boldsymbol{\theta}_0, 0)] = \mathbf{0}. \quad (\text{A.10})$$

Next, we need to ensure that F is continuously differentiable near $(\boldsymbol{\theta}_0, 0)$. Differentiability of F in $\boldsymbol{\theta}^*$ follows from conditions R2 and R5, which justify differentiating the score under the expectation. Differentiability of F in δ_0 follows from conditions R2 and R4, which justify differentiating the log-likelihood under the expectation with respect to the distributional parameter δ_0 . Finally, the Jacobian $\partial F / \partial \boldsymbol{\theta}^*$

$$\left. \frac{\partial F}{\partial \boldsymbol{\theta}^*} \right|_{(\boldsymbol{\theta}_0, 0)} = \mathbb{E}_{\delta_0=0}[\mathbf{H}_i^*(\boldsymbol{\theta}_0)] = -\mathcal{I}^*(\boldsymbol{\theta}_0), \quad (\text{A.11})$$

which is invertible at $(\boldsymbol{\theta}_0, 0)$ by condition R6. Therefore all conditions of the IFT are met, which implies the existence of a continuously differentiable function $\delta_0 \mapsto \boldsymbol{\theta}^*(\delta_0)$ defined in a neighbourhood of $\delta_0 = 0$ such that $F(\boldsymbol{\theta}^*(\delta_0), \delta_0) = \mathbf{0}$ and $\boldsymbol{\theta}^*(0) = \boldsymbol{\theta}_0$.

The second part of the lemma is necessary for the derivation of the nonlinear omitted variable bias. Since $\boldsymbol{\theta}^*(\delta_0)$ is continuously differentiable in a neighbourhood of $\delta_0 = 0$, we can obtain a first-order Taylor expansion,

$$\boldsymbol{\theta}^*(\delta_0) = \boldsymbol{\theta}_0 + \left. \frac{d\boldsymbol{\theta}^*}{d\delta_0} \right|_{\delta_0=0} \cdot \delta_0 + O(\delta_0^2). \quad (\text{A.12})$$

It remains to evaluate the derivative $d\boldsymbol{\theta}^*/d\delta_0$ at $\delta_0 = 0$. Totally differentiating the identity $F(\boldsymbol{\theta}^*(\delta_0), \delta_0) = \mathbf{0}$ with respect to δ_0 and rearranging:

$$\frac{d\boldsymbol{\theta}^*}{d\delta_0} = - \left(\frac{\partial F}{\partial \boldsymbol{\theta}^*} \right)^{-1} \frac{\partial F}{\partial \delta_0}. \quad (\text{A.13})$$

We know that $\partial F / \partial \boldsymbol{\theta}^* = -\mathcal{I}^*(\boldsymbol{\theta}_0)$ at $(\boldsymbol{\theta}_0, 0)$. It remains to evaluate $\partial F / \partial \delta_0$ at the same

point. Writing $F(\boldsymbol{\theta}^*, \delta_0) = \int \mathbf{s}^*(y, w; \boldsymbol{\theta}^*) p(y, w; \delta_0) dy dw$ and differentiating under the integral (justified by R4), the chain rule identity $\partial p / \partial \delta_0 = p \cdot s_{i,\delta}$ gives:

$$\frac{\partial F}{\partial \delta_0} = \mathbb{E}_{\delta_0}[\mathbf{s}_i^*(\boldsymbol{\theta}^*) s_{i,\delta}(\boldsymbol{\theta}_0, \delta_0)]. \quad (\text{A.14})$$

At $(\boldsymbol{\theta}_0, 0)$, the short-term score $\mathbf{s}_i^*(\boldsymbol{\theta}_0)$ has mean zero by (A.2), and $s_{i,\delta}(\boldsymbol{\theta}_0, 0)$ has mean zero by (A.5). Therefore,

$$\left. \frac{\partial F}{\partial \delta_0} \right|_{(\boldsymbol{\theta}_0, 0)} = \text{Cov}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)]. \quad (\text{A.15})$$

Substituting (A.15) and $\partial F / \partial \boldsymbol{\theta}^* = -\mathcal{I}^*(\boldsymbol{\theta}_0)$ into (A.13), the two minus signs cancel and:

$$\left. \frac{d\boldsymbol{\theta}^*}{d\delta_0} \right|_{\delta_0=0} = [\mathcal{I}^*(\boldsymbol{\theta}_0)]^{-1} \text{Cov}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)] := \boldsymbol{\Pi}. \quad (\text{A.16})$$

Substituting back into (A.12) yields (A.8). Since $\boldsymbol{\Pi}$ is a fixed finite matrix by R4 and R6, we obtain $\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*(\delta_0)\| = O(\delta_0)$ as $\delta_0 \rightarrow 0$.

Note that the covariance $\text{Cov}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)]$ is zero if and only if the short-model score at the truth is uncorrelated with the score of the omitted variable at $\delta_0 = 0$, i.e. Z_i carries no information about the short-model residuals at the truth. In that case $\boldsymbol{\Pi} = \mathbf{0}$ and $\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*(\delta_0)\| = O(\delta_0^2)$, so omitting Z_i causes no first-order distortion of the pseudo-true parameter. ■

The next lemma establishes that the Schur complement can be seen as a residualised covariance. This will allow us to give an interpretation to the nonlinear OVB similar to the linear OVB in Cinelli and Hazlett (2020).

Lemma A.2. *Let $a_i \in \mathbb{R}^p$ and $b_i \in \mathbb{R}^q$ be random vectors with finite second moments. Let $\mathcal{M}$ denote their joint $(p+q) \times (p+q)$ second-moment matrix, partitioned as:*

$$\mathcal{M} = \begin{pmatrix} \mathcal{M}_{AA} & \mathcal{M}_{AB} \\ \mathcal{M}_{BA} & \mathcal{M}_{BB} \end{pmatrix}, \quad (\text{A.17})$$

where $\mathcal{M}_{AA} = \mathbb{E}[a_i a_i'] \in \mathbb{R}^{p \times p}$, $\mathcal{M}_{AB} = \mathbb{E}[a_i b_i'] \in \mathbb{R}^{p \times q}$, $\mathcal{M}_{BB} = \mathbb{E}[b_i b_i'] \in \mathbb{R}^{q \times q}$, and $\mathcal{M}_{BA} = \mathcal{M}_{AB}' \in \mathbb{R}^{q \times p}$. Assume $\mathcal{M}_{AA}$ is invertible. Then the Schur complement of $\mathcal{M}_{AA}$ in $\mathcal{M}$,

$$\tilde{\mathcal{M}}_{BB} \equiv \mathcal{M}_{BB} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} \in \mathbb{R}^{q \times q}, \quad (\text{A.18})$$

equals the second-moment matrix of the projection residual $\tilde{b}_i = b_i - \Pi a_i \in \mathbb{R}^q$, where $\Pi = \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \in \mathbb{R}^{q \times p}$:

$$\tilde{\mathcal{M}}_{BB} = \mathbb{E}[\tilde{b}_i \tilde{b}_i']. \quad (\text{A.19})$$

If additionally $\mathbb{E}[b_i] = \Pi \mathbb{E}[a_i]$ (which holds in particular when $\mathbb{E}[a_i] = \mathbb{E}[b_i] = \mathbf{0}$, or when the projection includes a constant so that $\mathbb{E}[\tilde{b}_i] = \mathbf{0}$), then $\tilde{\mathcal{M}}_{BB} = \text{Cov}(\tilde{b}_i)$ is the covariance matrix of the residual.

Proof. Let Π be the coefficient matrix such that $b_i - \Pi a_i$ is orthogonal to a_i ,

$$\mathbb{E}[a_i (b_i - \Pi a_i)'] = \mathbf{0}. \quad (\text{A.20})$$

Expanding and substituting the definitions of $\mathcal{M}_{AB}$ and $\mathcal{M}_{AA}$:

$$\mathcal{M}_{AB} - \mathcal{M}_{AA} \Pi' = \mathbf{0} \implies \Pi' = \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} \implies \Pi = \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1}. \quad (\text{A.21})$$

So $\tilde{b}_i = b_i - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} a_i$ is the unique projection residual, satisfying:

$$\mathbb{E}[a_i \tilde{b}_i'] = \mathbf{0}. \quad (\text{A.22})$$

Now, expand the outer product

$$\begin{aligned} \mathbb{E}[\tilde{b}_i \tilde{b}_i'] &= \mathbb{E}[(b_i - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} a_i)(b_i - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} a_i)'] \\ &= \mathbb{E}[b_i b_i'] - \mathbb{E}[b_i a_i'] \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathbb{E}[a_i b_i'] \\ &\quad + \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathbb{E}[a_i a_i'] \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB}. \end{aligned} \quad (\text{A.23})$$

Substituting $\mathbb{E}[b_i b_i'] = \mathcal{M}_{BB}$, $\mathbb{E}[b_i a_i'] = \mathcal{M}_{BA}$, $\mathbb{E}[a_i b_i'] = \mathcal{M}_{AB}$, and $\mathbb{E}[a_i a_i'] = \mathcal{M}_{AA}$ into

(A.23):

$$\begin{aligned}\mathbb{E}[\tilde{b}_i \tilde{b}_i'] &= \mathcal{M}_{BB} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} + \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} \\ &= \mathcal{M}_{BB} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} = \tilde{\mathcal{M}}_{BB}.\end{aligned}\tag{A.24}$$

When $\mathbb{E}[\tilde{b}_i] = \mathbf{0}$ (e.g. if a_i includes an intercept), the second-moment matrix $\mathbb{E}[\tilde{b}_i \tilde{b}_i']$ equals the covariance matrix $\text{Cov}(\tilde{b}_i)$. ■

In our application, $a_i \in \mathbb{R}^{p+1}$ will be the nuisance covariate vector $A_i = (1, \mathbf{X}_i')' \in \mathbb{R}^{p+1}$ (or its w -weighted version $\sqrt{w_i} A_i$ in the GLM case), while the q -vector b_i is partitioned into two scalar components of interest, say $b_i = (D_i, Z_i)'$, so that $q = 2$. The blocks of $\mathcal{M}_{AB} \in \mathbb{R}^{p \times 2}$ and $\mathcal{M}_{BA} \in \mathbb{R}^{2 \times p}$ are:

$$\mathcal{M}_{AB} = \begin{pmatrix} \mathcal{M}_{AD} & \mathcal{M}_{AZ} \end{pmatrix}, \quad \mathcal{M}_{BA} = \begin{pmatrix} \mathcal{M}_{DA} \\ \mathcal{M}_{ZA} \end{pmatrix}, \tag{A.25}$$

where $\mathcal{M}_{AD} = \mathbb{E}[a_i D_i] \in \mathbb{R}^p$ and $\mathcal{M}_{AZ} = \mathbb{E}[a_i Z_i] \in \mathbb{R}^p$, and

$$\mathcal{M}_{BB} = \begin{pmatrix} \mathcal{M}_{DD} & \mathcal{M}_{DZ} \\ \mathcal{M}_{ZD} & \mathcal{M}_{ZZ} \end{pmatrix} \in \mathbb{R}^{2 \times 2}, \tag{A.26}$$

where $\mathcal{M}_{DD} = \mathbb{E}[D_i^2]$, $\mathcal{M}_{DZ} = \mathcal{M}_{ZD} = \mathbb{E}[D_i Z_i]$, and $\mathcal{M}_{ZZ} = \mathbb{E}[Z_i^2]$ are all scalars, whereas the Schur complement of $\mathcal{M}_{AA}$ in $\mathcal{M}$ is $\tilde{\mathcal{M}}_{BB} = \mathcal{M}_{BB} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} \in \mathbb{R}^{2 \times 2}$.

Substituting the partitioned forms into the product $\mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB}$ is:

$$\mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} = \begin{pmatrix} \mathcal{M}_{DA} \\ \mathcal{M}_{ZA} \end{pmatrix} \mathcal{M}_{AA}^{-1} \begin{pmatrix} \mathcal{M}_{AD} & \mathcal{M}_{AZ} \end{pmatrix}, \tag{A.27}$$

where $\mathcal{M}_{DA} \in \mathbb{R}^{1 \times p}$, $\mathcal{M}_{ZA} \in \mathbb{R}^{1 \times p}$, $\mathcal{M}_{AA}^{-1} \in \mathbb{R}^{p \times p}$, $\mathcal{M}_{AD} \in \mathbb{R}^{p \times 1}$, and $\mathcal{M}_{AZ} \in \mathbb{R}^{p \times 1}$. The

four scalar entries of the resulting 2×2 matrix are:

$$(1, 1) : \mathcal{M}_{DA}\mathcal{M}_{AA}^{-1}\mathcal{M}_{AD} \in \mathbb{R}, \quad (\text{A.28})$$

$$(1, 2) : \mathcal{M}_{DA}\mathcal{M}_{AA}^{-1}\mathcal{M}_{AZ} \in \mathbb{R}, \quad (\text{A.29})$$

$$(2, 1) : \mathcal{M}_{ZA}\mathcal{M}_{AA}^{-1}\mathcal{M}_{AD} \in \mathbb{R}, \quad (\text{A.30})$$

$$(2, 2) : \mathcal{M}_{ZA}\mathcal{M}_{AA}^{-1}\mathcal{M}_{AZ} \in \mathbb{R}. \quad (\text{A.31})$$

Subtracting block by block from $\mathcal{M}_{BB}$ gives the scalar entries of the 2×2 Schur complement $\tilde{\mathcal{M}}_{BB} = \mathcal{M}_{BB} - \mathcal{M}_{BA}\mathcal{M}_{AA}^{-1}\mathcal{M}_{AB}$ are:

$$\tilde{\mathcal{M}}_{DD} = \mathcal{M}_{DD} - \mathcal{M}_{DA}\mathcal{M}_{AA}^{-1}\mathcal{M}_{AD}, \quad (\text{A.32})$$

$$\tilde{\mathcal{M}}_{DZ} = \mathcal{M}_{DZ} - \mathcal{M}_{DA}\mathcal{M}_{AA}^{-1}\mathcal{M}_{AZ}, \quad (\text{A.33})$$

$$\tilde{\mathcal{M}}_{ZD} = \mathcal{M}_{ZD} - \mathcal{M}_{ZA}\mathcal{M}_{AA}^{-1}\mathcal{M}_{AD}, \quad (\text{A.34})$$

$$\tilde{\mathcal{M}}_{ZZ} = \mathcal{M}_{ZZ} - \mathcal{M}_{ZA}\mathcal{M}_{AA}^{-1}\mathcal{M}_{AZ}. \quad (\text{A.35})$$

Now, application of Lemma A.2 yields $\tilde{\mathcal{M}}_{DD} = \text{Var}(\tilde{D}_i)$, $\tilde{\mathcal{M}}_{ZZ} = \text{Var}(\tilde{Z}_i)$, and $\tilde{\mathcal{M}}_{DZ} = \text{Cov}(\tilde{D}_i, \tilde{Z}_i)$, where $\tilde{D}_i = D_i - \mathcal{M}_{DA}\mathcal{M}_{AA}^{-1}a_i$ and $\tilde{Z}_i = Z_i - \mathcal{M}_{ZA}\mathcal{M}_{AA}^{-1}a_i$ are the scalar residuals from projecting D_i and Z_i onto $a_i \in \mathbb{R}^{p+1}$. Since a_i includes an intercept (the first component of $A_i = (1, \mathbf{X}_i')'$), the orthogonality condition $\mathbb{E}[a_i\tilde{D}_i] = \mathbf{0}$ implies $\mathbb{E}[\tilde{D}_i] = 0$ and similarly $\mathbb{E}[\tilde{Z}_i] = 0$, so the second-moment matrices equal the covariance matrices.

B Proof of Lemma 2.1

Let conditions **R1**–**R6** hold. Partition the parameter vector into nuisance parameters $\boldsymbol{\theta}_A = (\alpha, \boldsymbol{\gamma}')' \in \mathbb{R}^{p+1}$ and parameters of interest $\boldsymbol{\theta}_B = (\beta, \delta)' \in \mathbb{R}^2$. The population information matrix evaluated at the pseudo-true parameter $\boldsymbol{\theta}^*$ partitions conformably as:

$$\mathcal{I}(\boldsymbol{\theta}^*) = \begin{pmatrix} \mathcal{I}_{AA}^* & \mathcal{I}_{AB}^* \\ \mathcal{I}_{BA}^* & \mathcal{I}_{BB}^* \end{pmatrix}, \quad (\text{B.1})$$

where $\mathcal{I}_{AA}^* \in \mathbb{R}^{(p+1) \times (p+1)}$, $\mathcal{I}_{AB}^* = (\mathcal{I}_{BA}^*)' \in \mathbb{R}^{(p+1) \times 2}$, and $\mathcal{I}_{BB}^* \in \mathbb{R}^{2 \times 2}$, with blocks $\mathcal{I}_{AA}^* = -\mathbb{E}[\mathbf{H}_{i,AA}(\boldsymbol{\theta}^*)]$, $\mathcal{I}_{AB}^* = -\mathbb{E}[\mathbf{H}_{i,AB}(\boldsymbol{\theta}^*)]$, and $\mathcal{I}_{BB}^* = -\mathbb{E}[\mathbf{H}_{i,BB}(\boldsymbol{\theta}^*)]$.

By score orthogonality (A.2), $\mathbf{s}(\boldsymbol{\theta}_0) = \mathbf{0}$. Taylor-expanding around $\boldsymbol{\theta}^*$ and substituting (A.1):

$$\mathcal{I}(\boldsymbol{\theta}^*)(\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*) = \mathbf{s}(\boldsymbol{\theta}^*) + O(\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|^2) = \mathbf{s}(\boldsymbol{\theta}^*) + O(\delta_0^2), \quad (\text{B.2})$$

where the second equality uses $\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\| = O(\delta_0)$ from Lemma A.1. Writing (B.2) in block form:

$$\begin{pmatrix} \mathcal{I}_{AA}^* & \mathcal{I}_{AB}^* \\ \mathcal{I}_{BA}^* & \mathcal{I}_{BB}^* \end{pmatrix} \begin{pmatrix} \boldsymbol{\theta}_{A,0} - \boldsymbol{\theta}_A^* \\ \boldsymbol{\theta}_{B,0} - \boldsymbol{\theta}_B^* \end{pmatrix} = \begin{pmatrix} \mathbf{s}_A(\boldsymbol{\theta}^*) \\ \mathbf{s}_B(\boldsymbol{\theta}^*) \end{pmatrix} + O(\delta_0^2). \quad (\text{B.3})$$

By (A.6), $\mathbf{s}_A(\boldsymbol{\theta}^*) = \mathbf{0}$. The top block of (B.3) is therefore:

$$\mathcal{I}_{AA}^*(\boldsymbol{\theta}_{A,0} - \boldsymbol{\theta}_A^*) + \mathcal{I}_{AB}^*(\boldsymbol{\theta}_{B,0} - \boldsymbol{\theta}_B^*) = O(\delta_0^2). \quad (\text{B.4})$$

Premultiplying by $(\mathcal{I}_{AA}^*)^{-1}$:

$$\boldsymbol{\theta}_{A,0} - \boldsymbol{\theta}_A^* = -(\mathcal{I}_{AA}^*)^{-1} \mathcal{I}_{AB}^*(\boldsymbol{\theta}_{B,0} - \boldsymbol{\theta}_B^*) + O(\delta_0^2). \quad (\text{B.5})$$

Substituting (B.5) into the second block of (B.3):

$$\begin{aligned}
\mathbf{s}_B(\boldsymbol{\theta}^*) + O(\delta_0^2) &= \mathcal{I}_{BA}^*(\boldsymbol{\theta}_{A,0} - \boldsymbol{\theta}_A^*) + \mathcal{I}_{BB}^*(\boldsymbol{\theta}_{B,0} - \boldsymbol{\theta}_B^*) \\
&= -\mathcal{I}_{BA}^*(\mathcal{I}_{AA}^*)^{-1}\mathcal{I}_{AB}^*(\boldsymbol{\theta}_{B,0} - \boldsymbol{\theta}_B^*) + \mathcal{I}_{BB}^*(\boldsymbol{\theta}_{B,0} - \boldsymbol{\theta}_B^*) + O(\delta_0^2) \\
&= \tilde{\mathcal{I}}_{BB}^*(\boldsymbol{\theta}_{B,0} - \boldsymbol{\theta}_B^*) + O(\delta_0^2),
\end{aligned} \tag{B.6}$$

where $\tilde{\mathcal{I}}_{BB}^* = \mathcal{I}_{BB}^* - \mathcal{I}_{BA}^*(\mathcal{I}_{AA}^*)^{-1}\mathcal{I}_{AB}^* \in \mathbb{R}^{2 \times 2}$ is the Schur complement of $\mathcal{I}_{AA}^*$ in $\mathcal{I}(\boldsymbol{\theta}^*)$. Since the short model restricts $\delta = 0$, we have $\delta^* = 0$ and $\boldsymbol{\theta}_{B,0} - \boldsymbol{\theta}_B^* = (\beta_0 - \beta^*, \delta_0)'$. By (A.6), $s_\beta(\boldsymbol{\theta}^*) = 0$, while $s_\delta(\boldsymbol{\theta}^*) \neq 0$ in general. Expanding the 2×2 system:

$$\begin{pmatrix} \tilde{\mathcal{I}}_{DD}^* & \tilde{\mathcal{I}}_{DZ}^* \\ \tilde{\mathcal{I}}_{ZD}^* & \tilde{\mathcal{I}}_{ZZ}^* \end{pmatrix} \begin{pmatrix} \beta_0 - \beta^* \\ \delta_0 \end{pmatrix} = \begin{pmatrix} 0 \\ s_\delta(\boldsymbol{\theta}^*) \end{pmatrix} + O(\delta_0^2). \tag{B.7}$$

The first scalar equation gives:

$$\tilde{\mathcal{I}}_{DD}^*(\beta_0 - \beta^*) + \tilde{\mathcal{I}}_{DZ}^* \delta_0 = O(\delta_0^2). \tag{B.8}$$

Solving for $\beta^* - \beta_0$:

$$\beta^* - \beta_0 = \frac{\tilde{\mathcal{I}}_{DZ}^*}{\tilde{\mathcal{I}}_{DD}^*} \delta_0 + O(\delta_0^2) = \tilde{\pi} \delta_0 + O(\delta_0^2), \tag{B.9}$$

where $\tilde{\pi} = \tilde{\mathcal{I}}_{DZ}^*/\tilde{\mathcal{I}}_{DD}^*$.

The ratio $\tilde{\pi} = \tilde{\mathcal{I}}_{DZ}^*/\tilde{\mathcal{I}}_{DD}^*$ has an exact interpretation as a ratio of projection residuals that follows directly from Lemma A.2, without any additional assumptions. Applying the lemma with $a_i = \mathbf{s}_{A,i}(\boldsymbol{\theta}^*)$ and $b_i = (s_{\beta,i}(\boldsymbol{\theta}^*), s_{\delta,i}(\boldsymbol{\theta}^*))'$, so that $\mathcal{M} = \mathcal{I}(\boldsymbol{\theta}^*)$, gives $\tilde{\mathcal{I}}_{DD}^* = \mathbb{E}[\tilde{s}_{\beta,i}^{*2}]$ and $\tilde{\mathcal{I}}_{DZ}^* = \mathbb{E}[\tilde{s}_{\beta,i}^* \tilde{s}_{\delta,i}^*]$, where $\tilde{s}_{\beta,i}^*$ and $\tilde{s}_{\delta,i}^*$ are the residuals from projecting the treatment score $s_{\beta,i}(\boldsymbol{\theta}^*)$ and the omitted-variable score $s_{\delta,i}(\boldsymbol{\theta}^*)$ onto the nuisance score vector $\mathbf{s}_{A,i}(\boldsymbol{\theta}^*)$ with respect to the second-moment inner product. Therefore, $\tilde{\pi}$ is the ratio of the projected cross-moment to the projected second moment of the treatment score. Note that this is a second-moment ratio rather than a covariance ratio, because $\mathbb{E}[\mathbf{s}_i(\boldsymbol{\theta}^*)] \neq \mathbf{0}$ in general when $\delta_0 \neq 0$: the short model is misspecified at $\boldsymbol{\theta}^*$ and the

information matrix equality does not hold there.

A population regression coefficient interpretation follows under the additional assumption that $\mathcal{I}(\boldsymbol{\theta})$ is continuously differentiable in $\boldsymbol{\theta}$ on Θ , which holds if the Hessian $\mathbf{H}_i(\boldsymbol{\theta})$ is once more differentiable with a suitable dominating function for its derivative (similar to R4 in an extended set of regularity conditions). Under this assumption, since $\boldsymbol{\theta}^* = \boldsymbol{\theta}_0 + O(\delta_0)$ by Lemma A.1, a first-order Taylor expansion gives $\mathcal{I}(\boldsymbol{\theta}^*) = \mathcal{I}(\boldsymbol{\theta}_0) + O(\delta_0)$, and hence $\tilde{\pi} = \tilde{\mathcal{I}}_{DZ}^0 / \tilde{\mathcal{I}}_{DD}^0 + O(\delta_0)$, where the superscript 0 denotes evaluation at $\boldsymbol{\theta}_0$. At $\boldsymbol{\theta}_0$ the model is correctly specified and the information matrix equality $\mathcal{I}(\boldsymbol{\theta}_0) = \mathbb{E}[\mathbf{s}_i(\boldsymbol{\theta}_0)\mathbf{s}_i(\boldsymbol{\theta}_0)']$ holds, so the second-moment matrices equal covariance matrices. Lemma A.2 then gives $\tilde{\mathcal{I}}_{DD}^0 = \text{Var}(\tilde{s}_{\beta,i}^0)$ and $\tilde{\mathcal{I}}_{DZ}^0 = \text{Cov}(\tilde{s}_{\beta,i}^0, \tilde{s}_{\delta,i}^0)$, where $\tilde{s}_{\beta,i}^0$ and $\tilde{s}_{\delta,i}^0$ are now residuals from projections with respect to the covariance inner product. It follows that in a neighbourhood of $\delta_0 = 0$, $\tilde{\pi}$ is approximately the population regression slope from regressing $\tilde{s}_{\delta,i}^0$ on $\tilde{s}_{\beta,i}^0$ after partialling out the nuisance scores.

For GLMs the structure of the Hessian permits a more transparent interpretation of $\tilde{\pi}$. For these models the Hessian takes the form $\mathbf{H}_i(\boldsymbol{\theta}) = -w_i(\boldsymbol{\theta})W_iW_i'$, where $w_i(\boldsymbol{\theta}) = V(\mu_i(\boldsymbol{\theta})) > 0$ is the variance weight and $W_i = (1, \mathbf{X}_i', D_i, Z_i)'$. The information matrix evaluated at the pseudo-true parameter is therefore

$$\mathcal{I}(\boldsymbol{\theta}^*) = \mathbb{E}[w_i^* W_i W_i'], \quad (\text{B.10})$$

where $w_i^* = V(\mu_i(\boldsymbol{\theta}^*))$, which is the second-moment matrix of $\sqrt{w_i^*}W_i$. This means we can apply Lemma A.2 a second time, now with

$$a_i = \sqrt{w_i^*} A_i \in \mathbb{R}^{p+1}, \quad b_i = \sqrt{w_i^*} (D_i, Z_i)' \in \mathbb{R}^2,$$

where $A_i = (1, \mathbf{X}_i')' \in \mathbb{R}^{p+1}$, so that $\mathcal{M} = \mathcal{I}(\boldsymbol{\theta}^*)$ is the full GLM information matrix partitioned by the split $(A_i, (D_i, Z_i)')$. The blocks are $\mathcal{I}_{AA}^* = \mathbb{E}[w_i^* A_i A_i'] \in \mathbb{R}^{(p+1) \times (p+1)}$ and

$$\mathcal{I}_{BB}^* = \begin{pmatrix} \mathcal{I}_{DD}^* & \mathcal{I}_{DZ}^* \\ \mathcal{I}_{ZD}^* & \mathcal{I}_{ZZ}^* \end{pmatrix} = \begin{pmatrix} \mathbb{E}[w_i^* D_i^2] & \mathbb{E}[w_i^* D_i Z_i] \\ \mathbb{E}[w_i^* Z_i D_i] & \mathbb{E}[w_i^* Z_i^2] \end{pmatrix} \in \mathbb{R}^{2 \times 2}, \quad (\text{B.11})$$

with off-diagonal block $\mathcal{I}_{AB}^* = \mathbb{E}[w_i^* A_i (D_i, Z_i)'] \in \mathbb{R}^{(p+1) \times 2}$. The cross-product $\mathcal{I}_{BA}^* (\mathcal{I}_{AA}^*)^{-1} \mathcal{I}_{AB}^*$ expands as

$$\mathcal{I}_{BA}^* (\mathcal{I}_{AA}^*)^{-1} \mathcal{I}_{AB}^* = \begin{pmatrix} \mathcal{I}_{DA}^* \\ \mathcal{I}_{ZA}^* \end{pmatrix} (\mathcal{I}_{AA}^*)^{-1} \begin{pmatrix} \mathcal{I}_{AD}^* & \mathcal{I}_{AZ}^* \end{pmatrix}, \quad (\text{B.12})$$

where $\mathcal{I}_{DA}^* = \mathbb{E}[w_i^* D_i A_i'] \in \mathbb{R}^{1 \times (p+1)}$, $\mathcal{I}_{ZA}^* = \mathbb{E}[w_i^* Z_i A_i'] \in \mathbb{R}^{1 \times (p+1)}$, $(\mathcal{I}_{AA}^*)^{-1} \in \mathbb{R}^{(p+1) \times (p+1)}$, $\mathcal{I}_{AD}^* = \mathbb{E}[w_i^* A_i D_i] \in \mathbb{R}^{(p+1) \times 1}$, and $\mathcal{I}_{AZ}^* = \mathbb{E}[w_i^* A_i Z_i] \in \mathbb{R}^{(p+1) \times 1}$. Subtracting block by block from $\mathcal{I}_{BB}^*$, the four scalar entries of $\tilde{\mathcal{I}}_{BB}^* = \mathcal{I}_{BB}^* - \mathcal{I}_{BA}^* (\mathcal{I}_{AA}^*)^{-1} \mathcal{I}_{AB}^*$ are:

$$\tilde{\mathcal{I}}_{DD}^* = \mathcal{I}_{DD}^* - \mathcal{I}_{DA}^* (\mathcal{I}_{AA}^*)^{-1} \mathcal{I}_{AD}^*, \quad (\text{B.13})$$

$$\tilde{\mathcal{I}}_{DZ}^* = \mathcal{I}_{DZ}^* - \mathcal{I}_{DA}^* (\mathcal{I}_{AA}^*)^{-1} \mathcal{I}_{AZ}^*, \quad (\text{B.14})$$

$$\tilde{\mathcal{I}}_{ZD}^* = \mathcal{I}_{ZD}^* - \mathcal{I}_{ZA}^* (\mathcal{I}_{AA}^*)^{-1} \mathcal{I}_{AD}^*, \quad (\text{B.15})$$

$$\tilde{\mathcal{I}}_{ZZ}^* = \mathcal{I}_{ZZ}^* - \mathcal{I}_{ZA}^* (\mathcal{I}_{AA}^*)^{-1} \mathcal{I}_{AZ}^*. \quad (\text{B.16})$$

Now applying Lemma A.2 to the weighted second-moment matrix, the Schur complement entries equal the second moments of the weighted projection residuals:

$$\tilde{\mathcal{I}}_{DD}^* = \mathbb{E}[w_i^* \tilde{D}_i^{*2}], \quad \tilde{\mathcal{I}}_{DZ}^* = \mathbb{E}[w_i^* \tilde{D}_i^* \tilde{Z}_i^*], \quad (\text{B.17})$$

where

$$\begin{aligned} \tilde{D}_i^* &= D_i - (\boldsymbol{\pi}_D^*)' A_i, & \boldsymbol{\pi}_D^* &= \mathbb{E}[w_i^* A_i A_i']^{-1} \mathbb{E}[w_i^* A_i D_i], \\ \tilde{Z}_i^* &= Z_i - (\boldsymbol{\pi}_Z^*)' A_i, & \boldsymbol{\pi}_Z^* &= \mathbb{E}[w_i^* A_i A_i']^{-1} \mathbb{E}[w_i^* A_i Z_i], \end{aligned}$$

are the residuals from w^* -weighted projections of D_i and Z_i onto A_i . Therefore,

$$\tilde{\pi} = \frac{\mathbb{E}[w_i^* \tilde{D}_i^* \tilde{Z}_i^*]}{\mathbb{E}[w_i^* \tilde{D}_i^{*2}]}, \quad (\text{B.18})$$

which is a w^* -weighted ratio of second moments of the covariate residuals $\tilde{D}_i^*$ and $\tilde{Z}_i^*$. As in the general case, this is an exact second-moment ratio rather than a covariance

ratio, because $\mathbb{E}[\tilde{D}_i^*] \neq 0$ and $\mathbb{E}[\tilde{Z}_i^*] \neq 0$ in general when $\delta_0 \neq 0$: the weights w_i^* are evaluated at the misspecified pseudo-true parameter and the weighted projections need not be mean-zero.

Under the same additional differentiability assumption invoked in the general case, $w_i^* = w_i^0 + O(\delta_0)$ where $w_i^0 = V(\mu_i(\boldsymbol{\theta}_0))$, and (B.18) becomes

$$\tilde{\pi} = \frac{\text{Cov}_{w^0}(\tilde{D}_i^0, \tilde{Z}_i^0)}{\text{Var}_{w^0}(\tilde{D}_i^0)} + O(\delta_0), \quad (\text{B.19})$$

where $\tilde{D}_i^0$ and $\tilde{Z}_i^0$ are the residuals from w^0 -weighted projections of D_i and Z_i onto A_i , and Cov_{w^0} and Var_{w^0} denote w^0 -weighted covariance and variance. Since A_i includes an intercept, the orthogonality condition $\mathbb{E}[w_i^0 A_i \tilde{D}_i^0] = \mathbf{0}$ implies $\mathbb{E}_{w^0}[\tilde{D}_i^0] = 0$ and similarly $\mathbb{E}_{w^0}[\tilde{Z}_i^0] = 0$, so the second-moment matrices equal the covariance matrices at $\boldsymbol{\theta}_0$. As a result, in a neighbourhood of $\delta_0 = 0$, $\tilde{\pi}$ is approximately the population regression slope from regressing $\tilde{Z}_i^0$ on $\tilde{D}_i^0$ using weights w_i^0 . That is a nonlinear analogue of the OLS treatment residual slope in the linear OVB formula of Cinelli and Hazlett (2020).

C Proof of Proposition 3.1

Let $\tilde{D}^*$ (with $\mathbb{E}[\tilde{D}^*] = 0$ and $\text{Var}(\tilde{D}^*) = 1$), $\tilde{Y}^*$ (with $\mathbb{E}[\tilde{Y}^*] = 0$ and $\text{Var}(\tilde{Y}^*) = 1$) and let $\text{Cov}(\tilde{D}^*, \tilde{Y}^*) = r$ (and hence $\text{Cor}(\tilde{D}^*, \tilde{Y}^*) = r$). The variable $\varepsilon \sim N(0, 1)$ independent of $(\tilde{D}^*, \tilde{Y}^*)$. Define the pseduo-confounder

$$Z = a \cdot \tilde{D}^* + b \cdot \tilde{Y}^* + c \cdot \varepsilon \quad (\text{C.1})$$

We find a, b, c such that $\text{Cor}(Z, \tilde{D}^*) = \rho_D$ and $\text{Cor}(Z, \tilde{Y}^*) = \rho_Y$.

Noting that $\mathbb{E}[Z] = 0$, it follows that

$$\begin{aligned} \text{Cov}(Z, \tilde{D}^*) &= \text{Cov}(a\tilde{D}^* + b\tilde{Y}^* + c\varepsilon, \tilde{D}^*) \\ &= a \cdot \text{Cov}(\tilde{D}^*, \tilde{D}^*) + b \cdot \text{Cov}(\tilde{Y}^*, \tilde{D}^*) + c \cdot \text{Cov}(\varepsilon, \tilde{D}^*) \\ &= a + br \end{aligned} \quad (\text{C.2})$$

$$\begin{aligned} \text{Cov}(Z, \tilde{Y}^*) &= \text{Cov}(a\tilde{D}^* + b\tilde{Y}^* + c\varepsilon, \tilde{Y}^*) \\ &= ar + b \end{aligned} \quad (\text{C.3})$$

and

$$\begin{aligned} \text{Var}(Z) &= \text{Var}(a\tilde{D}^* + b\tilde{Y}^* + c\varepsilon) \\ &= a^2\text{Var}(\tilde{D}^*) + b^2\text{Var}(\tilde{Y}^*) + c^2\text{Var}(\varepsilon) \\ &\quad + 2ab\text{Cov}(\tilde{D}^*, \tilde{Y}^*) + 2ac\text{Cov}(\tilde{D}^*, \varepsilon) + 2bc\text{Cov}(\tilde{Y}^*, \varepsilon) \\ &= a^2 + b^2 + c^2 + 2abr \end{aligned} \quad (\text{C.4})$$

Next, we can express correlations in terms of covariances. The correlation between Z and $\tilde{D}^*$ is:

$$\text{Cor}(Z, \tilde{D}^*) = \frac{\text{Cov}(Z, \tilde{D}^*)}{\sqrt{\text{Var}(Z)} \cdot \sqrt{\text{Var}(\tilde{D}^*)}} = \frac{a + br}{\sqrt{a^2 + b^2 + c^2 + 2abr}} \quad (\text{C.5})$$

$$\text{Cor}(Z, \tilde{Y}^*) = \frac{ar + b}{\sqrt{a^2 + b^2 + c^2 + 2abr}} \quad (\text{C.6})$$

Without loss of generality, we normalise Z to have unit variance.

$$a^2 + b^2 + c^2 + 2abr = 1 \quad \Rightarrow \quad c^2 = 1 - a^2 - b^2 - 2abr \quad (\text{C.7})$$

With $\text{Var}(Z) = 1$, the correlations simplify to:

$$\text{Cor}(Z, \tilde{D}^*) = a + br \quad (\text{C.8})$$

$$\text{Cor}(Z, \tilde{Y}^*) = ar + b \quad (\text{C.9})$$

Next, we solve for a and b . We require:

$$a + br = \rho_D \quad (\text{C.10})$$

$$ar + b = \rho_Y \quad (\text{C.11})$$

This is a system of two linear equations in two unknowns (a and b). Assuming $r^2 \neq 1$ (i.e., $\tilde{D}^*$ and $\tilde{Y}^*$ are not perfectly correlated)¹⁷:

$$b = \frac{\rho_Y - \rho_D r}{1 - r^2} \quad (\text{C.12})$$

$$a = \frac{\rho_D - \rho_Y r}{1 - r^2} \quad (\text{C.13})$$

From the normalization condition $c^2 = 1 - a^2 - b^2 - 2abr$, for a real solution, we need $c^2 \geq 0$. Define $u = \rho_D - \rho_Y r$ and $v = \rho_Y - \rho_D r$ so that $a = u/(1 - r^2)$ and $b = v/(1 - r^2)$.

Then,

$$a^2 + b^2 + 2abr = \frac{u^2 + v^2 + 2uvr}{(1 - r^2)^2} \quad (\text{C.14})$$

¹⁷To verify that result, just note, for example, that

$$\begin{aligned} a + br &= \frac{\rho_D - \rho_Y r}{1 - r^2} + \frac{\rho_Y - \rho_D r}{1 - r^2} \cdot r = \frac{\rho_D - \rho_Y r + (\rho_Y - \rho_D r)r}{1 - r^2} \\ &= \frac{\rho_D - \rho_Y r + \rho_Y r - \rho_D r^2}{1 - r^2} = \frac{\rho_D(1 - r^2)}{1 - r^2} = \rho_D \end{aligned}$$

Furthermore,

$$u^2 = \rho_D^2 - 2\rho_D\rho_Yr + \rho_Y^2r^2 \quad (\text{C.15})$$

$$v^2 = \rho_Y^2 - 2\rho_D\rho_Yr + \rho_D^2r^2 \quad (\text{C.16})$$

and

$$\begin{aligned} u^2 + v^2 &= \rho_D^2 + \rho_Y^2r^2 + \rho_Y^2 + \rho_D^2r^2 - 4\rho_D\rho_Yr \\ &= (\rho_D^2 + \rho_Y^2)(1 + r^2) - 4\rho_D\rho_Yr \end{aligned} \quad (\text{C.17})$$

$$\begin{aligned} uv &= (\rho_D - \rho_Yr)(\rho_Y - \rho_Dr) \\ &= \rho_D\rho_Y - \rho_D^2r - \rho_Y^2r + \rho_D\rho_Yr^2 \\ &= \rho_D\rho_Y(1 + r^2) - r(\rho_D^2 + \rho_Y^2) \end{aligned} \quad (\text{C.18})$$

It follows that

$$\begin{aligned} u^2 + v^2 + 2uvr &= (\rho_D^2 + \rho_Y^2)(1 + r^2) - 4\rho_D\rho_Yr + 2r[\rho_D\rho_Y(1 + r^2) - r(\rho_D^2 + \rho_Y^2)] \\ &= (\rho_D^2 + \rho_Y^2)(1 + r^2) - 4\rho_D\rho_Yr + 2\rho_D\rho_Yr(1 + r^2) - 2r^2(\rho_D^2 + \rho_Y^2) \\ &= (\rho_D^2 + \rho_Y^2)(1 + r^2 - 2r^2) + \rho_D\rho_Yr(-4 + 2 + 2r^2) \\ &= (\rho_D^2 + \rho_Y^2)(1 - r^2) + 2\rho_D\rho_Yr(r^2 - 1) \\ &= (\rho_D^2 + \rho_Y^2)(1 - r^2) - 2\rho_D\rho_Yr(1 - r^2) \\ &= (1 - r^2)(\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Yr) \end{aligned} \quad (\text{C.19})$$

Therefore:

$$a^2 + b^2 + 2abr = \frac{(1 - r^2)(\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Yr)}{(1 - r^2)^2} = \frac{\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Yr}{1 - r^2} \quad (\text{C.20})$$

The feasibility condition $c^2 \geq 0$ becomes:

$$1 - \frac{\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Yr}{1 - r^2} = \frac{(1 - r^2) - (\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Yr)}{1 - r^2} \geq 0 \quad (\text{C.21})$$

Since $1 - r^2 > 0$ (assuming $|r| < 1$), this is equivalent to:

$$\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Yr \leq 1 - r^2 \tag{C.22}$$

□

D Supporting Results

Table 9: Partial Correlations of Age-10 Cognitive Skills (PC1) with Covariates

Variable	Partial corr.	Semipartial corr.	Partial corr. ²	<i>p</i> -value
<i>Outcomes</i>				
Employed full-time	0.0427	0.0343	0.0018	0.038
Manager or supervisor	0.0628	0.0505	0.0039	0.002
<i>Non-cognitive skills</i>				
Perseverance (age 10, SD)	0.2807	0.2346	0.0788	0.000
<i>Birth and perinatal characteristics (Set 1)</i>				
Mother's age at delivery	−0.0003	−0.0003	0.0000	0.987
Mother's age at delivery ²	0.0120	0.0096	0.0001	0.560
Region 1	0.0230	0.0184	0.0005	0.266
Region 2	0.0219	0.0176	0.0005	0.289
Region 3	−0.0006	−0.0005	0.0000	0.975
Region 4		(dropped)		
Region 5	0.0172	0.0138	0.0003	0.404
Region 6	−0.0109	−0.0088	0.0001	0.596
Region 7	0.0050	0.0040	0.0000	0.810
Region 8	0.0158	0.0126	0.0002	0.445
Region 9	−0.0233	−0.0187	0.0005	0.260
Region 10	0.0221	0.0178	0.0005	0.283
Mother smoked during pregnancy	−0.0289	−0.0232	0.0008	0.162
Abnormal gestation	−0.0310	−0.0249	0.0010	0.133
Female	−0.0575	−0.0462	0.0033	0.005
Birth weight	0.0431	0.0346	0.0019	0.037
<i>Age-5 family background (Set 2)</i>				

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Table 9 continued from previous page

Variable	Partial corr.	Semipartial corr.	Partial corr. ²	p-value
Mother stayed in education	0.0719	0.0579	0.0052	0.001
Father stayed in education	0.0561	0.0451	0.0031	0.007
Parity	−0.0684	−0.0550	0.0047	0.001
Previous stillbirth	0.0346	0.0278	0.0012	0.093
Previous preterm birth	−0.0075	−0.0060	0.0001	0.717
Previous stillbirth (high)	0.0100	0.0080	0.0001	0.627
Previous miscarriages	−0.0022	−0.0018	0.0000	0.914
Breastfed	0.0416	0.0334	0.0017	0.044
<i>Age-5 health (Set 3)</i>				
Inpatient stay	0.0080	0.0064	0.0001	0.699
Hearing or sight problems	−0.0219	−0.0176	0.0005	0.288
Child health rating	0.0194	0.0156	0.0004	0.347
<i>Parental education (Set 3 cont.)</i>				
Mother: non-level qualification	−0.0572	−0.0460	0.0033	0.006
Mother: secondary qualification	−0.0330	−0.0265	0.0011	0.110
Mother: degree	0.0412	0.0330	0.0017	0.046
Father: non-level qualification	−0.0260	−0.0209	0.0007	0.208
Father: secondary qualification	0.0072	0.0058	0.0001	0.727
Father: degree	0.0351	0.0282	0.0012	0.089
<i>Maternal employment and household (Set 3 cont.)</i>				
Mother employed full-time (age 5)	−0.0120	−0.0096	0.0001	0.561
Mother employed part-time (age 5)	−0.0020	−0.0016	0.0000	0.922
Non-white	−0.0409	−0.0329	0.0017	0.047
No siblings at age 5	−0.0019	−0.0015	0.0000	0.927

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Table 9 continued from previous page

Variable	Partial corr.	Semipartial corr.	Partial corr. ²	p-value
<i>Age-10 health and home environment (Set 4)</i>				
Child health (age 10)	−0.0103	−0.0082	0.0001	0.619
Inpatient stay (age 10)	0.0157	0.0126	0.0002	0.447
Father unemployed (age 10)	−0.0131	−0.0105	0.0002	0.527
Father stay-at-home (age 10)	−0.0206	−0.0165	0.0004	0.319
Mother unemployed (age 10)	−0.0194	−0.0156	0.0004	0.347
Mother stay-at-home (age 10)	−0.0035	−0.0028	0.0000	0.867
Benefit receipt	0.0276	0.0221	0.0008	0.181
Income band 1	0.0317	0.0255	0.0010	0.124
Income band 2		(dropped)		
Income band 3	0.0454	0.0364	0.0021	0.028
Income band 4	0.0426	0.0342	0.0018	0.039
Income band 5	0.0288	0.0231	0.0008	0.162
Income band 6	0.0557	0.0448	0.0031	0.007
Income band 7	0.0480	0.0386	0.0023	0.020
<i>Accommodation type (Set 5)</i>				
Accommodation type 1	−0.0220	−0.0177	0.0005	0.286
Accommodation type 2	−0.0202	−0.0162	0.0004	0.328
Accommodation type 3	−0.0268	−0.0215	0.0007	0.194
Accommodation type 4	−0.0250	−0.0201	0.0006	0.225
Accommodation type 5		(dropped)		
Accommodation type 6	−0.0227	−0.0182	0.0005	0.271
Accommodation type 7	−0.0161	−0.0129	0.0003	0.437
Number of rooms	0.0241	0.0194	0.0006	0.242
<i>Parental job satisfaction (Set 6)</i>				

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Table 9 continued from previous page

Variable	Partial corr.	Semipartial corr.	Partial corr. ²	<i>p</i> -value
Mother's job satisfaction	−0.0155	−0.0124	0.0002	0.454
Father's job satisfaction	−0.0210	−0.0169	0.0004	0.308
<i>Medical and sensory (Set 6 cont.)</i>				
Eyesight problems	−0.0002	−0.0002	0.0000	0.992
Hearing problems	−0.0396	−0.0318	0.0016	0.055
Speech problems	0.0257	0.0207	0.0007	0.212
<i>Physical development</i>				
Height (adjusted)	0.1095	0.0884	0.0120	0.000
Weight (adjusted)	0.1082	0.0874	0.0117	0.000
<i>Teacher questionnaire and classroom behaviour PCs (Set 7)</i>				
Perserverance (age 10)	−0.0329	−0.0264	0.0011	0.111
Behaviour PC B	0.1536	0.1247	0.0236	0.000
Behaviour PC D	−0.0423	−0.0340	0.0018	0.040
Behaviour PC E	−0.0028	−0.0022	0.0000	0.893
Behaviour PC F	−0.0436	−0.0350	0.0019	0.034
Behaviour PC G	−0.0045	−0.0036	0.0000	0.829
Behaviour PC H	−0.0036	−0.0029	0.0000	0.860

Notes: $N = 2,422$. Partial and semipartial correlations from `pcorr` in Stata, conditioning on the full covariate set. PC1 is the first principal component of five age-10 cognitive test scores. *p*-values are two-tailed. Variables marked (dropped) were collinear with other regressors.